

Unit-3.

Applications of OP-AMP ~

2 types:

1. Linear application. [linear relationship btw i/p & o/p]
2. Non-Linear application.

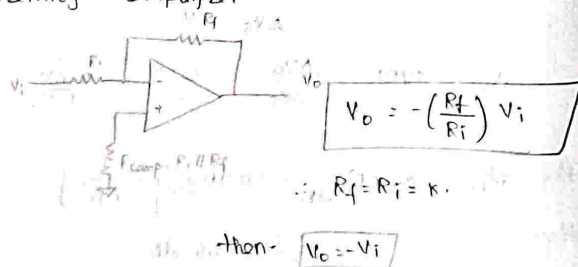
Linear

- current to voltage converter.
- voltage to current converter.
- adder.
- subtractor.
- summing amplifier.
- instrumentation amplifier.

Non-Linear

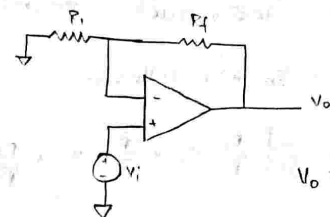
- Differentiator.
- Integrator.
- Rectifier.
- low pass filter.
- high pass filter.
- peak detector.
- logarithmic amplifier.

①. Inverting amplifier.



$V_o = -V_i \rightarrow$ act as scale changer / inverter.

②. Non-inverting amplifier.

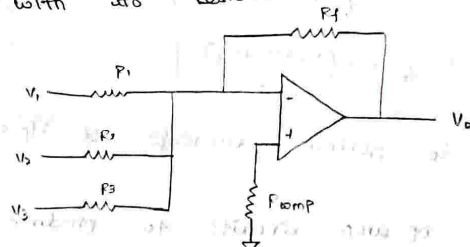


$$V_o = \left[1 + \frac{R_f}{R_i}\right] V_i$$

Inverting summing amplifier.

$\Rightarrow$ to get this, we modify the inverting amplifier.

* By applying multiple inputs along with its resistance.



* It performs addition and amplification.

$$V_{o1} \Rightarrow V_o \text{ due to } 'V_1' \text{ alone} = -\frac{R_f}{R_1} V_1$$

$$V_{o2} \Rightarrow \text{o/p } (V_o) \text{ due to } 'V_2' \text{ alone} = -\frac{R_f}{R_2} V_2$$

$$V_{o3} \Rightarrow V_o \text{ due to } 'V_3' \text{ alone} = -\frac{R_f}{R_3} V_3$$

by superposition theorem,

Total output is equal to output of each individual source.

$$V_o = V_{o1} + V_{o2} + V_{o3}.$$

$$V_o = - \left[\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} V_3 \right]$$

if $R_1 = R_2 = R_3 = R_f$

then, $V_o = -(V_1 + V_2 + V_3)$ It performs addition of i/p's.

It has 180° phase shift
btw i/p & o/p.

if $R_1 = R_2 = R_3 = 3R_f$.

$$V_o = - \left[\frac{V_1}{3} + \frac{V_2}{3} + \frac{V_3}{3} \right]$$

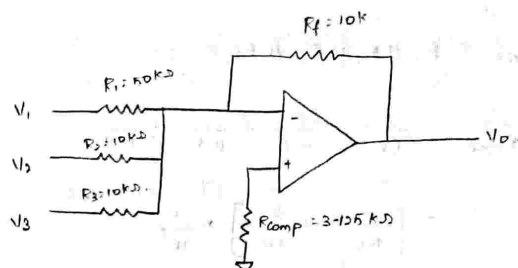
$$V_o = - \frac{(V_1 + V_2 + V_3)}{3}$$

It can able to perform average of i/p's.

① design an op-amp circuits to produce the output expression of $V_o = -(0.2V_1 + V_2 + V_3)$

Sol: Inverting summing amplifier.

$$V_o = - \left[\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} V_3 \right]$$



compare equation,

$$\frac{R_f}{R_1} V_1 = 0.2 V_1$$

$$\frac{R_f}{R_1} = 0.2$$

$$R_1 = \frac{R_f}{0.2}$$

Let choose $R_f = 10k$.

$$R_1 = \frac{10k}{0.2} = 50k\Omega$$

Similarly,

$$\frac{R_f}{R_2} V_2 = V_2$$

$$R_f = R_2$$

$$\therefore R_2 = 10k\Omega$$

$$\frac{R_f}{R_3} V_3 = V_3$$

$$R_3 = R_f$$

$$\therefore R_3 = 10k\Omega$$

$$R_{comp} = R_1 \parallel R_2 \parallel R_3 \parallel R_f$$

$$\frac{1}{R_{comp}} = \frac{1}{50k} + \frac{1}{10k} + \frac{1}{10k} + \frac{1}{10k}$$

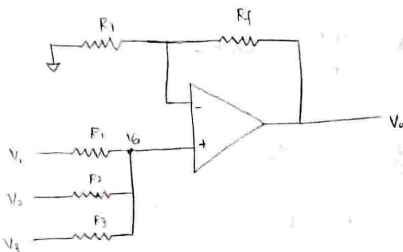
$$= \left[\frac{1}{50} + \frac{3}{10} \right] \times \frac{1}{10^3}$$

$$\frac{1}{R_{comp}} = \frac{16}{50} \times \frac{1}{10^3}$$

$$R_{comp} = \frac{50}{16} \times 10^3$$

$$R_{comp} = 3.125 \times 10^3 \Omega$$

Non Inverting - Summing Amplifier.



$$V_o = \left[1 + \frac{R_f}{R_1} \right] V_a \rightarrow (1)$$

Apply nodal analysis at V_a .

$$\frac{V_a - V_1}{R_1} + \frac{V_a - V_2}{R_2} + \frac{V_a - V_3}{R_3} = 0$$

$$V_o \left[\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right] = \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3}$$

$$\therefore V_o = \frac{\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3}}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}} \rightarrow (2)$$

Substitute (1) in eqn (2).

$$V_o = \left[1 + \frac{R_f}{R_1} \right] \frac{\left[\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \right]}{\left[\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right]}$$

Let us assume that

$$R_1 = R_2 = R_3 = R$$

$$V_o = \left[1 + \frac{R_f}{R_1} \right] \frac{V_1 + V_2 + V_3}{\frac{3}{R}}$$

$$V_o = \left(1 + \frac{R_f}{R_1} \right) \left(\frac{V_1 + V_2 + V_3}{3} \right)$$

choose, $R_f = \frac{R_1}{2}$

$$V_o = \left[1 + \frac{R_f}{R_1} \right] \left[\frac{V_1 + V_2 + V_3}{3} \right]$$

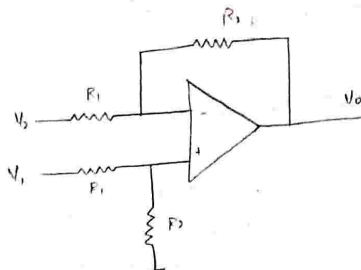
$$V_o = (2) \frac{V_1 + V_2 + V_3}{3}$$

$$\therefore V_o = V_1 + V_2 + V_3$$

$\therefore$ input and output of circuit

is same phase.

Subtractor:
A Difference Amplifier is used here.



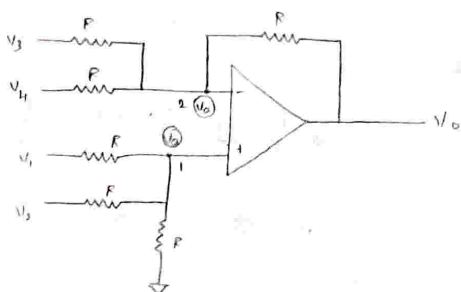
$$V_0 = \frac{R_3}{R_1} (V_1 - V_2)$$

if $R_3 = R_1$,

$$\text{then } V_0 = (V_1 - V_2)$$

$\Rightarrow$ It performs subtractor.

Adder - subtractor circuit:



Apply nodal analysis at node 1,

$$\frac{V_0 - V_1}{R} + \frac{V_0 - V_2}{R} + \frac{V_0 - 0}{R} = 0$$

$$\therefore V_0 \left[\frac{1}{R} + \frac{1}{R} + \frac{1}{R} \right] = \frac{V_1}{R} + \frac{V_2}{R}$$

$$V_0 = \frac{V_1 + V_2}{\frac{3}{R}}$$

$$V_0 \left[\frac{3}{R} \right] = \frac{V_1 + V_2}{R} \rightarrow \text{①}$$

Apply nodal analysis at node 2,

$$\frac{V_0 - V_3}{R} + \frac{V_0 - V_4}{R} + \frac{V_0 - V_5}{R} = 0$$

$$V_0 \left[\frac{1}{R} + \frac{1}{R} + \frac{1}{R} \right] = \frac{V_3}{R} + \frac{V_4}{R} + \frac{V_5}{R}$$

$$V_0 \left[\frac{3}{R} \right] = \frac{V_3 + V_4 + V_5}{R} \rightarrow \text{②}$$

Equate ① & ②.

$$\frac{V_1 + V_2}{R} = \frac{V_3 + V_4 + V_5}{R}$$

$$\therefore V_0 = (V_1 + V_2) - (V_3 + V_4)$$

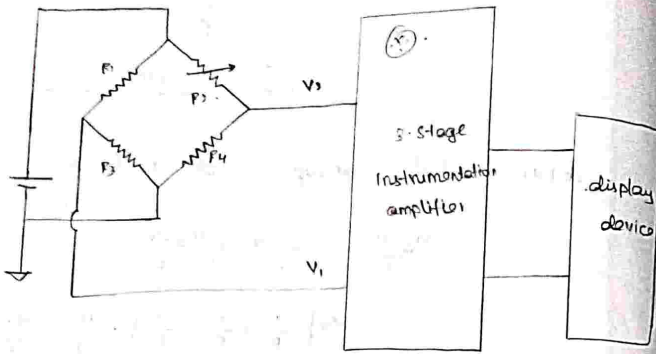
Instrumentation amplifier: [transducer bridge Based Instrumentation amplifier]

Transducer - convert non-electrical signal to electrical signal.

It is able to measure any physical quantity.

eg temp, humidity, water flow, pressure

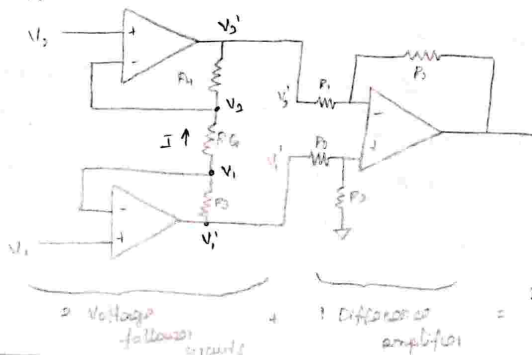
The o/p of transducer is very weak. It is connected to amplifier. In that we can see the o/p.



V_2 & V_1 is weak so it connected to instrumentation amplifier.

* In difference amplifier loading effect is created becoz the input impedance are low to amplify.

* For instrumentation amplitude, the input sources given through the voltage follower circuit to the difference amplifier, by this loading effect is overcome.



Advantage:

1. It offers high gain accuracy.
2. It offers high CMRR.
3. offers high stability.
4. low offset voltages.
5. low output impedance.

$$V_0 = \frac{R_2}{R_1} (V_1' - V_2') \rightarrow (1)$$

here $I = \frac{V_1 - V_2}{R_4} \Rightarrow V_1 - V_2 = IR_4 \rightarrow (2)$

no current flows thru input terminal

so, R_3, R_1, R_4 are connected in series

$$I = \frac{V_1' - V_2'}{R_3 + R_1 + R_4}$$

Let, $R_3 = R_4$

$$I = \frac{V_1' - V_2'}{2R_3 + R_4} \rightarrow (3)$$

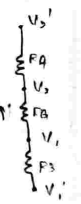
equate (1) & (3).

$$\frac{V_1 - V_2}{R_4} = \frac{V_1' - V_2'}{2R_3 + R_4}$$

$$\therefore V_1' - V_2' = \frac{2R_3 + R_4}{R_4} (V_1 - V_2)$$

$$V_1' - V_2' = \left[1 + \frac{2R_3}{R_4} \right] (V_1 - V_2) \rightarrow (4)$$

substitute (4) in eqn (1).



$$V_o = \left(\frac{R_3}{R_1} \right) \left(1 + \frac{R_3}{R_4} \right) (V_1 - V_2)$$

$\frac{R_3}{R_1} \rightarrow$ gain of difference amplifier.

$1 + \frac{R_3}{R_4} \rightarrow$ gain of $\rightarrow$ voltage follower amplifiers.

Voltage to Current Converter:

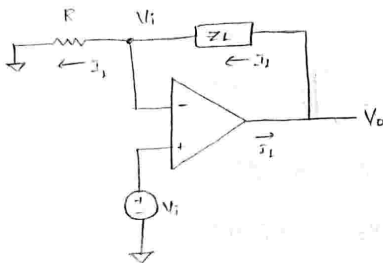
[Transconductance amplifier]

1. Floating load type V-I converter

2. Grounded load type V-I converter.

$$\Rightarrow \text{gain} = \frac{I}{V} = \frac{1}{R} = g_m$$

1. Floating load type V-I converter.



Load impedance is not grounded, it is floating, so connected to o/p.

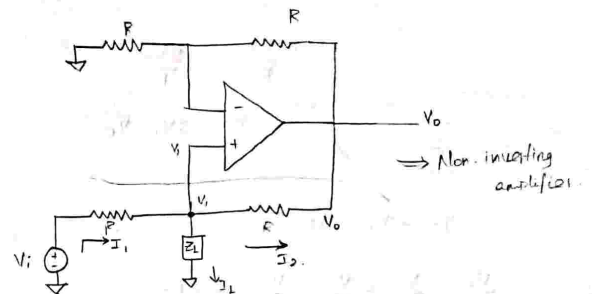
The applied voltage should decide the current.

$$I_1 = \frac{V_1}{R} \Rightarrow V_1 = I_1 R$$

Precision resistor:

At any cost, value of impedance won't change.

2. Grounded load type V-I converter.



Here the load impedance is connected to ground.

Two steps are followed.

1. Apply KCL w.r.t V_1 node.
2. Non-inverting amplifier.

$\therefore$ o/p of Non-inverting amplifier when V_1 is known.

$$V_o = \left[1 + \frac{R}{R} \right] V_1$$

$$V_o = 2V_1 \rightarrow \text{①}$$

Apply KCL eqn w.r.t to Node ' V_1 '

$$I_2 + I_1 = I_L$$

$$\frac{V_0 - V_i}{R} + \frac{V_i - V_i}{R} = \frac{V_i - 0}{Z_L}$$

Let us assume $Z_L = R$.

$$\frac{V_0 - V_i}{R} + \frac{V_i - V_i}{R} = \frac{V_i}{R}$$

$$\frac{V_0 + V_i}{R} = \frac{2V_i}{R}$$

$$V_i = 2V_i - V_0$$

$$I_2 + I_1 = I_L$$

$$\frac{V_0}{R} - \frac{V_i}{R} + \frac{V_i}{R} - \frac{V_i}{R} = I_L$$

$$\frac{V_0 + V_i}{R} - \frac{2V_i}{R} = I_L$$

$$V_0 = I_L R + 2V_i - V_i \rightarrow \textcircled{1}$$

Sub $\textcircled{1}$ in $\textcircled{2}$.

$$V_0 = I_L R + \frac{2V_0}{2} - V_i$$

$$V_i = I_L R$$

$$I_L = \frac{V_i}{R}$$

Precision resistor

V-I:

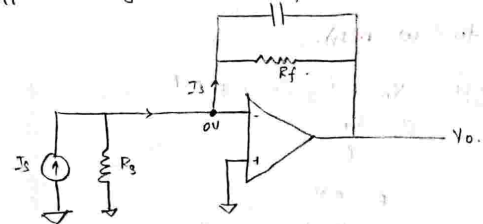
* current produced with respect to voltage.

circuit, which produces constant current

through load impedance.

Current to Voltage Converter:

Applied input current will get o/p voltage.



assume, there is no feedback capacitance.

$\therefore C_f \rightarrow$ eliminate high frequency noise signals.

apply KCL at in node.

$$I_s = \frac{0 - V_o}{R_f}$$

$$V_o = -I_s R_f$$

$$\text{gain} \Rightarrow \frac{V_o}{I_s} = -R_f = R_f$$

Gain of circuit expressed in resistance

$\therefore$ It is also known as transresistance amplifier.

C_f is not needed when we using low frequency signals.

- ③ Design a V-I converter with grounded load using an ideal op-amp such that i/p voltage range (0 to 5)V, output current range (0 to 10 mA).

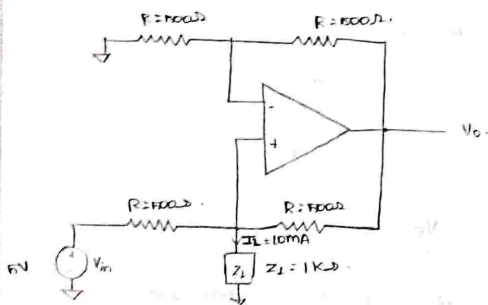
sol: Given, $V_{in} = 5V$, $I_L = 10mA$.

$$I_L = \frac{V_{in}}{R}$$

$$R = \frac{5V}{10 \times 10^{-3}}$$

$$R = 0.5 K\Omega$$

$$R = 500 \Omega$$



choose $Z_L = 1K\Omega$.

- ③ Design a grounded load V-I converter that produces o/p current is 2 mA/V of input signal, and maximum o/p voltage is 4 V.

sol:

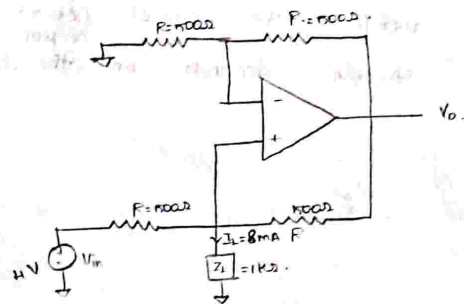
$$I_L = \frac{V_{in}}{R}$$

$$\frac{2mA}{1V} = \frac{I_L}{V_{in}} = \frac{1}{R}$$

$$R = \frac{1}{2 \times 10^{-3}}$$

$$R = 0.5 K\Omega$$

$$R = 500\Omega$$



$$\text{when } V_{in} = 1V \rightarrow I_L = 2mA$$

$$V_{in} = 4V \rightarrow I_L = 4 \times 2 \times 10^{-3}$$

$$I_L = 8mA$$

Resistance remains same. (that's why it is precision resistance).

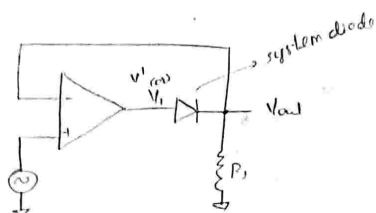
⇒ when we use the input less than cut-off voltage of diode ($< 0.7V$), the rectifier can't rectify. then we can go for op-amp based rectifier circuit which performs rectification.

Super diode:

whenever the combination of two diodes and op-amp.

Precision Rectifier:

Due to its open loop gain, the diode rectifies the signal ($< 0.7V$) ($< \mu V$) V_{out} strongly depends on the diode



$$V_1 = A(V_{in} - V_{out}) \quad (\text{or}) \quad V_1 = A(V_+ - V_-)$$

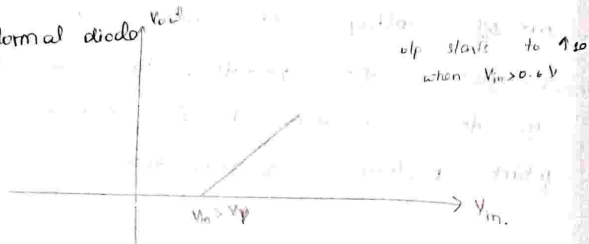
$$V_1 = 0.7 + V_{out}$$

$$0.7 + V_{out} = A(V_{in} - V_{out})$$

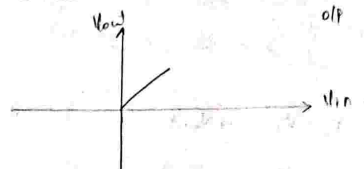
$$\frac{0.7 + V_{out}}{A} = V_{in} - V_{out}$$

for ideal diode $\rightarrow$ No knee voltage & built-in voltage.

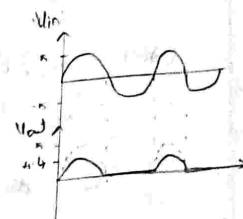
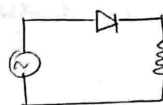
1) Normal diode



2) Super diode:



o/p comes at 0.0 itself.



for non-ideal diode

* Usually, diode has internal built-in voltage of $0.6V$ - Silicon, $0.3V$ - Germanium.

This diode is non-ideal diode.

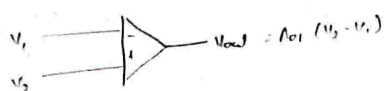
* It gives the o/p by reducing $0.6V$. It doesn't rectify when i/p is less than $0.6V$.

* In graph, it reduces the amplitude, whereas in ideal there is no change in amplitude.

* If we want o/p for $(i/p < 0.6V)$ then we use op-amp with diode.

$$\text{So, } V_{out} = \frac{0.6}{A_{ol}} = \frac{0.6}{10^4} = 60 \mu V.$$

* Now at $60 \times 10^{-6} V$ the diode can rectify.



$$V_2 > V_1 \Rightarrow +V_{out}$$

$$V_2 < V_1 \Rightarrow -V_{out}$$

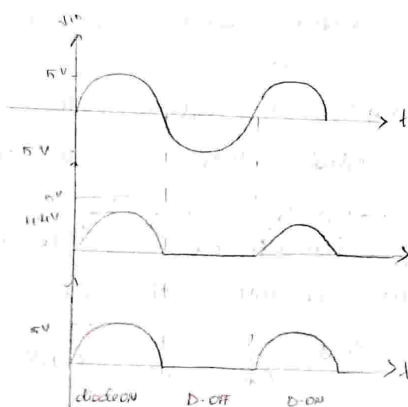
when, $V_1 < 0$.

$$V_1 = -V_{sat}$$

The diode is off & reverse bias

$$V_o = 0$$

In precision rectifier, it perform rectification with no drop voltage, in output.



Limitation:

*. It is only suitable to perform for low frequency signal. (due to it's slow rate it may not get o/p)

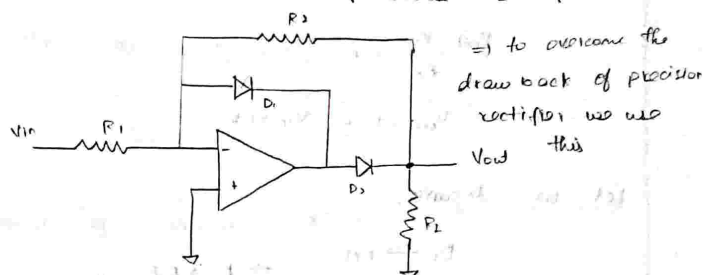
$V_{in} \rightarrow$ during +ve half cycle.

$$V_o' = +V_{sat}$$

$V_{in} \rightarrow$ during -ve half cycle.

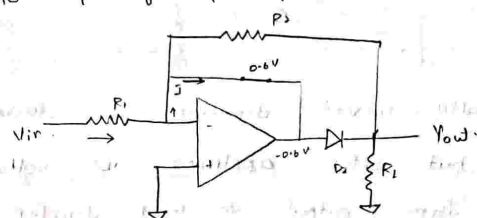
$$V_o' = -V_{sat}$$

modified precision half wave rectifier.



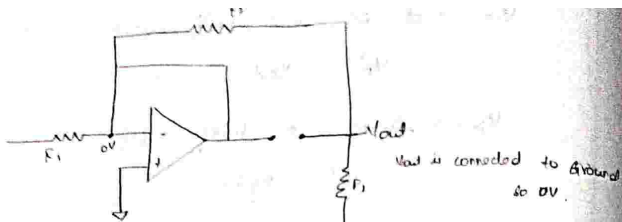
Let as assume, $D_1 \rightarrow ON$, when $V_{in} > 0V$.

D_1 & D_2 is not directly connected to o/p of op-amp.



$D_2 \rightarrow OFF$ [beac -0.6V to the terminal connected so reverse bias occurs].

Due to virtual ground concept -ve terminal has 0V. The circuit, $V_{out} = 0$.



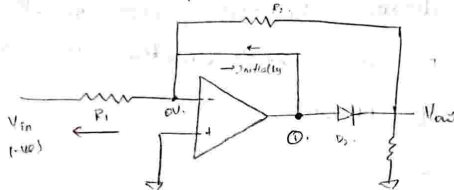
$$\frac{V_{out} - V_{in}}{R_2} = 0$$

$$V_{out} = 0, \quad V_{in} > 0V$$

Let us assume,

$D_1 \rightarrow ON, \Rightarrow D_1 \rightarrow R.B$

$V_{in} = -10V$



Initially current direction is towards \$V_{out}\$.

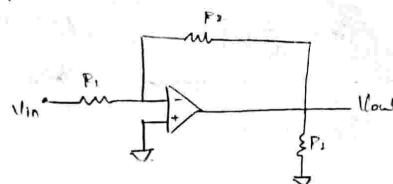
D-F.B but after applying -ve voltage I flows from output to input direction, direction changes, so \$D_1\$ is R.B, finally \$D_1 \rightarrow OFF\$.

$\therefore$ Voltage at 1 is greater than 0V (\$V_{in}\$)

so +ve is connected to +ve of \$D_2\$.

\$D_2\$ - F.B, \$D_2 \rightarrow ON\$.

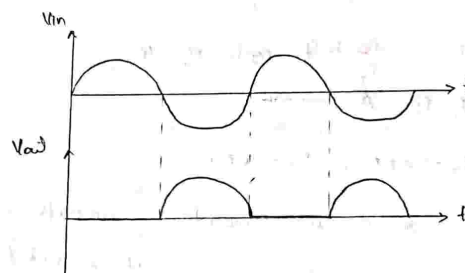
If \$V_{in} = -10V\$, it is Inverting amplifier.



$$V_{out} = \left(-\frac{R_2}{R_1} \right) (-V_{in})$$

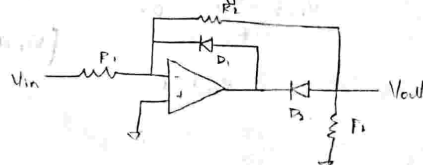
$$V_{out} = \frac{R_2}{R_1} V_{in}$$

Example of modified precision Rectifier circuit.



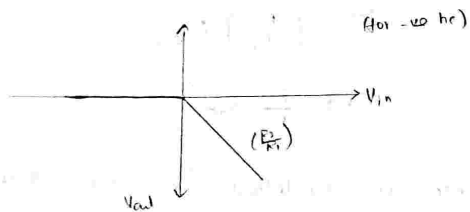
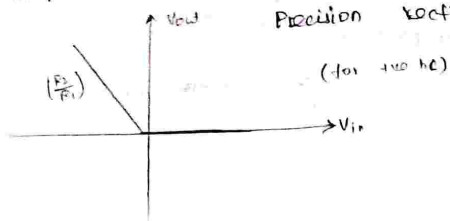
* only +ve half of circuit is rectified.

* If we want to rectify -ve half cycle, we change the bias of \$D_1, D_2\$.



(It rectify +ve & -ve half cycle)

Input & output characteristics of Precision Rectifier.



$V_{in} > 0$, +ve half cycle of i/p.

o/p of 'A1' $\rightarrow -V_e$.

$D_1 \rightarrow \text{OFF}$, $D_2 \rightarrow \text{ON}$.

It is inverting amplifier.

$$V_o = -\frac{R_2}{R_1} V_{in} \quad \text{It is rectified with amplitude } \frac{R_2}{R_1}$$

$V_{in} < 0$, -ve half cycle of i/p.

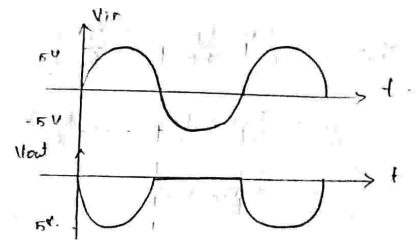
o/p of 'A1' $\rightarrow +V_e$.

$D_1 \rightarrow \text{ON}$, $D_2 \rightarrow \text{OFF}$.

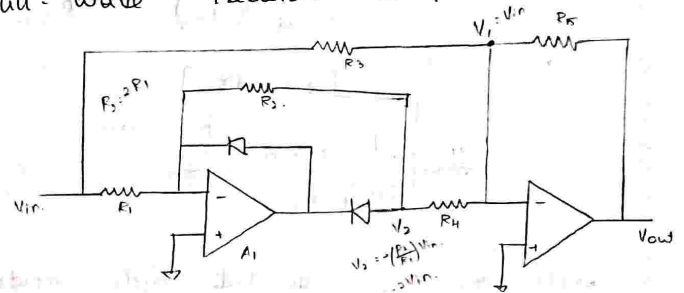
$$\frac{V_{out} - V_{in}}{R_2} = 0$$

$$V_{out} = 0$$

$$[V_{in} = 0]$$



Full-wave Precision Rectifier.



It is the combination modified half-wave rectifier followed by inverting summing amplifier.

When $V_{in} > 0$, [+ve half cycle i/p]

o/p of 'A1' $\rightarrow -V_e$.

$D_1 \rightarrow \text{OFF}$ (R.B)

$D_2 \rightarrow \text{ON}$ (F.B).

$$V_o = V_s = -\frac{R_2}{R_1} V_{in} \quad (-ve)$$

$$V_i = V_{in} = +ve$$

When, $V_i = V_s$ with opp polarity.

$$R_1 = R_2 = R_4 = R$$

$$V_{ad} = - \left(V_1 \frac{R_5}{R_4} + V_2 \frac{R_5}{R_3} \right)$$

$$= - \left(-V_{in} \frac{R_5}{R_4} + V_{in} \frac{R_5}{R_3} \right)$$

$$= - (-V_{in} + V_{in})$$

$$= 0$$

$(R_5 = R_4 = R_3 = R)$

$$V_{out} = - \left[-V_{in} \left(\frac{R_5}{R_4} \right) + V_{in} \left(\frac{R_5}{R_3} \right) \right]$$

$$= - [-V_{in} + V_{in}]$$

$$= - (-V_{in})$$

$$\boxed{V_{out} = V_{in}}$$

when $V_{in} < 0$, -ve half cycle input.

o/p of 'A₁' $\rightarrow +V_{in}$.

$D_1 \rightarrow ON$, $D_2 \rightarrow OFF$

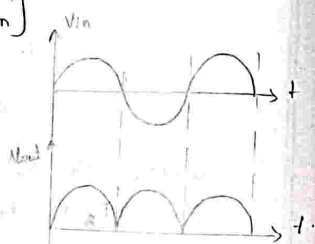
$V_{out} = 0$.

$\therefore V_2 = 0$, $V_1 = -V_{in}$.

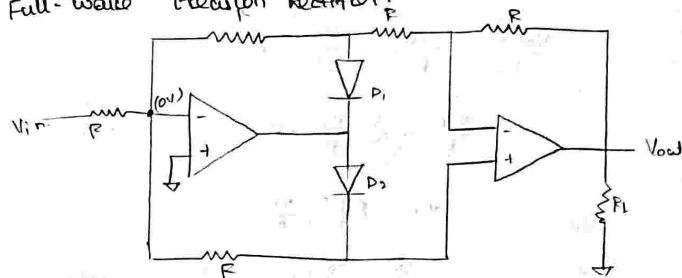
$$V_o = - \left[V_2 \left(\frac{R_5}{R_4} \right) + V_1 \left(\frac{R_5}{R_3} \right) \right]$$

$$= - [0 - V_{in}]$$

$$V_o = +V_{in}$$



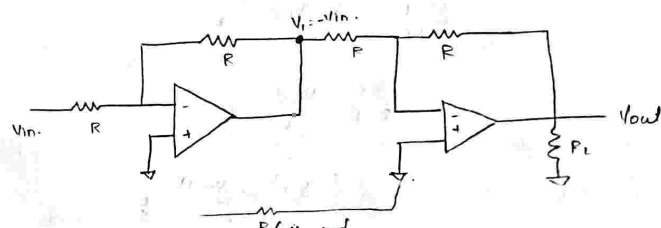
Full-wave Precision Rectifier:



$V_{in} > 0$, +ve half cycle of i/p.

$D_1 \rightarrow ON$ (short circuit)

$D_2 \rightarrow OFF$ (open circuit).



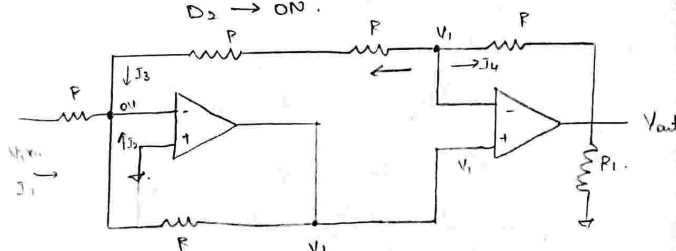
$$V_1 = V_{o1} = -V_{in}$$

$$V_{out} = -(-V_{in}) = V_{in}$$

$V_{in} < 0$, -ve half cycle of i/p.

$D_1 \rightarrow OFF$

$D_2 \rightarrow ON$.



$$\frac{V_{in}-0}{R} + \frac{V_1-0}{R} + \frac{V_1}{2R} = 0.$$

$$\frac{V_{in}}{R} + \frac{3}{2} \frac{V_1}{R} = 0.$$

$$V_{in} = -R \left(\frac{3}{2} \right) \frac{V_1}{R}.$$

$$\boxed{V_{in} = -\frac{3}{2} V_1}$$

$$V_1 = -\frac{2}{3} V_{in}.$$

$$I_3 + I_4 = 0.$$

$$\frac{V_1-0}{2R} + \frac{V_1-V_{out}}{R} = 0.$$

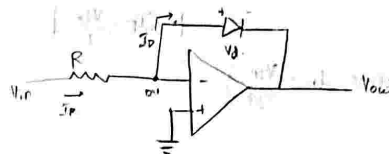
$$\frac{3V_1}{2R} - \frac{V_{out}}{R} = 0.$$

$$\frac{3}{2} \left(-\frac{2}{3} \frac{V_{in}}{R} \right) - \frac{V_{out}}{R} = 0.$$

$$\frac{-V_{in}}{R} = \frac{V_{out}}{R}.$$

$$\boxed{V_{out} = -V_{in}.$$

Logarithmic Amplifier



$I_S \rightarrow$ reverse saturation current

$I_D \rightarrow$ diode current

$V_D \rightarrow$ diode voltage

$V_T \rightarrow$ Volt equivalent temperature

$$\text{diode current equation } I_D = I_S \left[e^{V_D/\eta V_T} - 1 \right]$$

①

where,

$$V_T = \frac{kT}{q}$$

$k \rightarrow$ boltzman's constant $= 1.38 \times 10^{-23} \text{ J/K}$

$q \rightarrow 1.602 \times 10^{-19} \text{ C}$ [coulomb]

$T \rightarrow$ temperature.

$$V_T = \frac{kT}{q} \approx 26 \text{ mV at room temp (25}^\circ\text{C)}$$

from the given circuit.

$$0 + V_D + V_{out} = 0.$$

$$V_{out} = -V_D. \rightarrow \text{②}$$

$$\text{From equ ①, } \frac{I_D}{I_S} = \left[e^{V_D/\eta V_T} - 1 \right]$$

$$\frac{I_D}{I_S} + 1 = e^{V_D/\eta V_T}.$$

$$\therefore \frac{I_D}{I_S} \gg 1, \text{ neglect } 1.$$

$$\frac{I_D}{I_S} = e^{V_D/\eta V_T}$$

apply ln on b/s.

$$\ln \left(\frac{I_D}{I_S} \right) = \ln \left(e^{V_D/\eta V_T} \right)$$

$$V_D = \eta V_T \ln \left(\frac{I_P}{I_S} \right)$$

$$[\because I_P = \frac{V_{in}}{R}]$$

$$V_D = \eta V_T \ln \left(\frac{V_{in}}{I_S R} \right)$$

from equ ①,

$$V_{out} = -V_D$$

$$V_{out} = -\eta V_T \ln \left(\frac{V_{in}}{I_S R} \right)$$

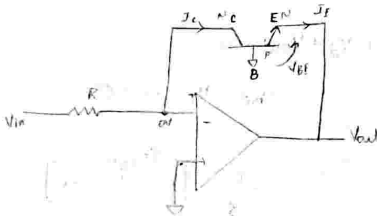
output $\propto$ logarithm of V_{in}

V_{out} depends on η , we don't need

this so we constructed transistor base logarithm

op-amp drawback: presence of η , dependence of temperature parameter

Transistor Base Log - amplifier:



$$I_E = I_C + I_B \quad [\because \text{Base is grounded } I_B = 0]$$

$$I_E \approx I_C$$

collector current,

$$I_C = I_S \left[e^{V_E/V_T} - 1 \right] \rightarrow ①$$

from the circuit.

$$0 + V_E + V_{out} = 0$$

$$V_{out} = -V_E \rightarrow ②$$

from equ ①,

$$\frac{I_C}{I_S} = \left[e^{V_E/V_T} - 1 \right]$$

$$\frac{I_C}{I_S} + 1 = e^{V_E/V_T}$$

$$\because \frac{I_C}{I_S} \gg 1, 1 \text{ is neglected.}$$

$$\frac{I_C}{I_S} = e^{V_E/V_T}$$

apply natural log on b.s.

$$\frac{V_E}{V_T} = \ln \left(\frac{I_C}{I_S} \right)$$

$$V_E \xrightarrow{I_P} 0$$

$$\frac{V_E}{V_T} = \ln \left(\frac{I_P}{I_S} \right)$$

$$[\because I_P = \frac{V_{in}}{R}]$$

$$V_E = V_T \ln \left(\frac{V_{in}}{I_S R} \right) \rightarrow ③$$

sub ③ in ①.

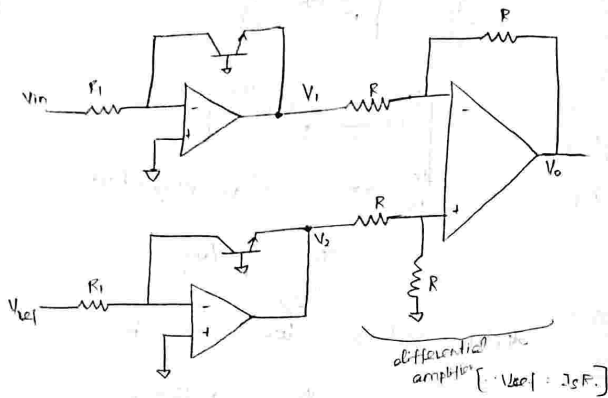
$$V_{out} = -V_T \ln \left(\frac{V_{in}}{I_S R} \right)$$

$$V_{out} = -\frac{kT}{q} \ln \left(\frac{V_{in}}{I_S R} \right)$$

Drawback:

Here, still V_{out} is dependent of temperature.

Temperature compensated transistor based logarithmic amplifier.



$$V_1 = -\frac{KT}{q} \ln \left[\frac{V_{in}}{I_S R_1} \right]$$

$$V_2 = -\frac{KT}{q} \ln \left[\frac{V_{ref}}{I_S R_1} \right]$$

By differential amplifier we know,

$$V_o = +\frac{R_2}{R_1} (V_2 - V_1)$$

$$\therefore V_{out} = \frac{R}{R} (V_2 - V_1)$$

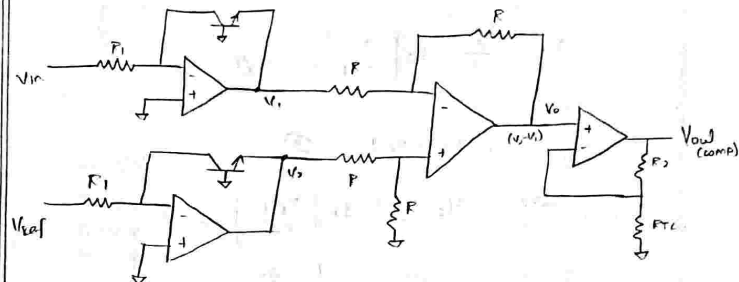
$$V_{out} = -\frac{KT}{q} \ln \left(\frac{V_{ref}}{I_S R_1} \right) + \frac{KT}{q} \ln \left(\frac{V_{in}}{I_S R_1} \right)$$

$$V_{out} = \frac{KT}{q} \ln \left[\frac{V_{in}}{V_{ref}} \right]$$

In order to compensate the T

we append one more circuit, as non-inverting

amplifier.

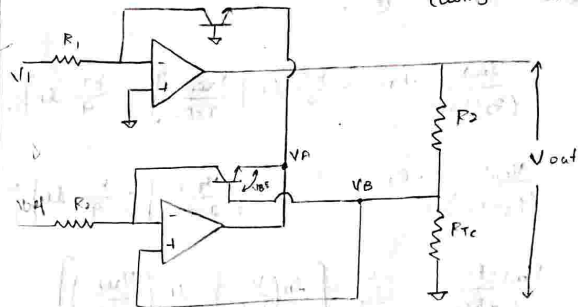


$R_{TC} \rightarrow$ temperature dependent resistor

$$V_{out(Comp)} = \left(1 + \frac{R_2}{R_{TC}} \right) \frac{KT}{q} \ln \left[\frac{V_{in}}{V_{ref}} \right]$$

when $T \uparrow$ so R_{TC} also $\uparrow$ so both cancel each other, so temperature effect is cancelled.

Temperature compensated log-amplifier: (using $\propto$ log-amp).



V_A - collector terminal of Transistor 1 &
collector terminal of Transistor 2.

$$V_A = -\frac{KT}{q} \ln \left[\frac{V_1}{I_S R_1} \right] \rightarrow (1)$$

$$\therefore V_B - V_E = V_A \rightarrow (2)$$

here, $V_E = \frac{KT}{q} \ln \left[\frac{V_{ref}}{I_S R_1} \right] \Rightarrow$ derived from transistor base log-amplifier

$\rightarrow (3)$

sub (3) in (2).

$$V_B - \frac{KT}{q} \ln \left[\frac{V_{ref}}{I_S R_1} \right] = -\frac{KT}{q} \ln \left[\frac{V_1}{I_S R_1} \right] \rightarrow (4)$$

here V_B = voltage across R_{TC}

by voltage divider theorem, $V_B = \frac{V_{out} \cdot R_{TC}}{(R_3 + R_{TC})}$

$\rightarrow (5)$

sub (5) in (4),

$$\frac{V_{out}}{(R_3 + R_{TC})} \cdot R_{TC} - \frac{KT}{q} \ln \left[\frac{V_{ref}}{I_S R_1} \right] = -\frac{KT}{q} \ln \left[\frac{V_1}{I_S R_1} \right]$$

$$\frac{V_{out}}{(R_3 + R_{TC})} \cdot R_{TC} = \frac{KT}{q} \ln \left[\frac{V_{ref}}{I_S R_1} \right] - \frac{KT}{q} \ln \left[\frac{V_1}{I_S R_1} \right]$$

$$\frac{V_{out} \cdot R_{TC}}{R_3 + R_{TC}} = -\frac{KT}{q} \left[\ln \left(\frac{V_1}{I_S R_1} \right) - \ln \left(\frac{V_{ref}}{I_S R_1} \right) \right]$$

V_B depends upon V_{out}

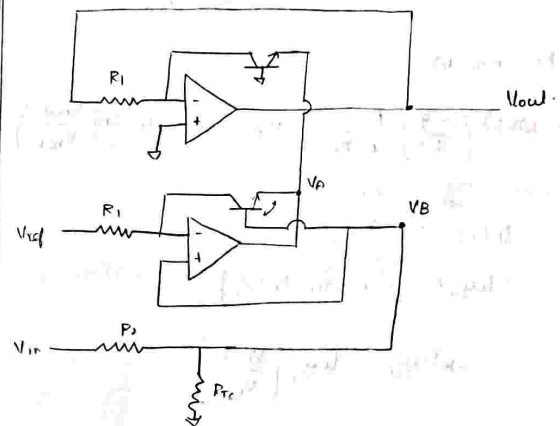
$$V_{out} = \left(\frac{R_3 + R_{TC}}{R_{TC}} \right) \left(-\frac{KT}{q} \right) \ln \left[\frac{\frac{V_1}{I_S R_1}}{\frac{V_{ref}}{I_S R_1}} \right]$$

$$\therefore V_{out} = \left[1 + \frac{R_3}{R_{TC}} \right] \left(-\frac{KT}{q} \right) \ln \left(\frac{V_1}{V_{ref}} \right)$$

This is how we can design a temperature compensated log-amplifier using only two logarithmic op-amp.

eg $\log(1000) + 3$
 $\text{antilog}(10^3) = 1000$

Anti-log amplifier:



$$V_A = -\frac{KT}{q} \ln \left[\frac{V_{out}}{I_S R_1} \right] \rightarrow (6)$$

$$V_B - V_E = V_A$$

$$V_B = \frac{KT}{q} \ln \left[\frac{V_{ref}}{I_S R_1} \right] = -\frac{KT}{q} \ln \left[\frac{V_{out}}{I_S R_1} \right]$$

$$\frac{V_{in} \cdot R_{TC}}{R_3 + R_{TC}} = \frac{KT}{q} \ln \left[\frac{V_{ref}}{I_S R_1} \right] = -\frac{KT}{q} \ln \left[\frac{V_{out}}{I_S R_1} \right]$$

$$\frac{V_{in} \cdot R_{TC}}{R_3 + R_{TC}} = \frac{KT}{q} \ln \left[\frac{V_{ref}}{I_S R_1} \right] - \frac{KT}{q} \ln \left[\frac{V_{out}}{I_S R_1} \right]$$

$$\frac{V_{in} R_{TC}}{R_3 + R_{TC}} = -\frac{KT}{q} \left[\ln \left(\frac{V_{out}}{I_S R_1} \right) - \ln \left(\frac{V_{ref}}{I_S R_1} \right) \right]$$

$$\left(\frac{-q}{KT} \right) \left(\frac{R_{TC}}{R_3 + R_{TC}} \right) V_{in} = \ln \left(\frac{V_{out}}{V_{ref}} \right)$$

xy by 0.4343,

$$(0.4343) \left(\frac{-q}{KT} \right) \frac{R_{TC}}{R_3 + R_{TC}} \cdot V_{in} = 0.4343 \ln \left(\frac{V_{out}}{V_{ref}} \right)$$

constants replace by k'

$$\therefore \ln(x) = 2.3 \log_{10}(x)$$

$$\log_{10}(x) = 0.4343 \ln(x)$$

$$-k' V_{in} = \log_{10} \left[\frac{V_{out}}{V_{ref}} \right]$$

$$\therefore \frac{V_{out}}{V_{ref}} = 10^{-k' V_{in}}$$

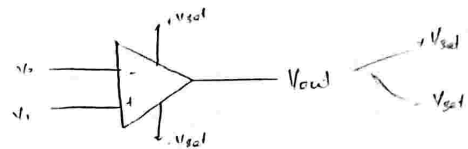
$$\therefore V_{out} = V_{ref} 10^{-k' V_{in}}$$

eg: If $k' = V_{in} = 1$ then

$$V_{out} = \frac{V_{ref}}{10}$$

V_{ref} is by 10 times.

COMPARATOR:



$$V_{out} = A_{OL} (v_1 - v_2) \quad [\because \text{If } (v_1 - v_2) \text{ is small}]$$

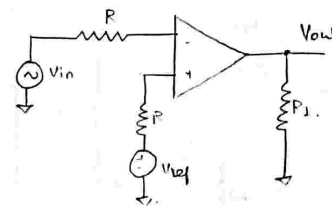
$$= +V_{sat} \text{ (or)} -V_{sat} \quad \text{due } A_{OL} (10^6) \text{ we get this}$$

It compare two terminal voltage, higher voltage terminal, the o/p go to corresponding V_{sat} .

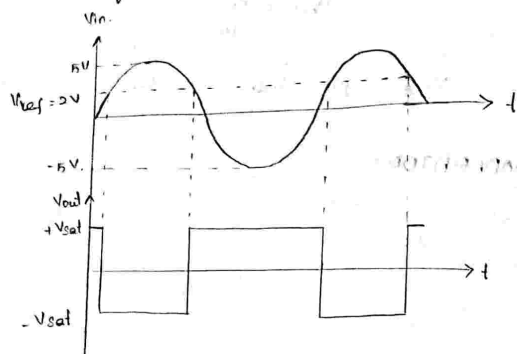
when, $V_1 > V_2$, o/p: $+V_{sat}$

$V_2 > V_1$, o/p: $-V_{sat}$

Inverting mode comparator:



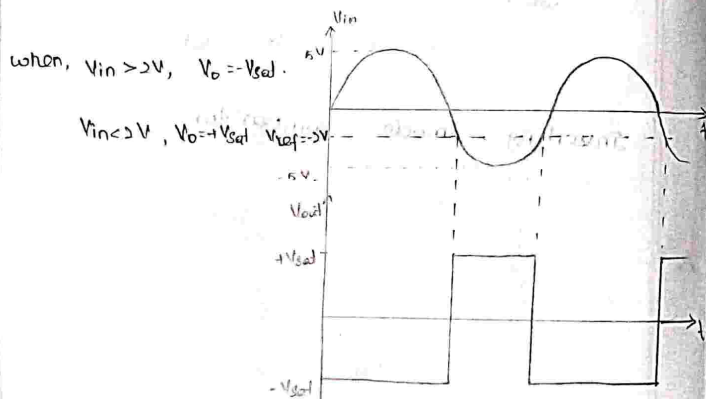
Assume, $V_{ref} = +V$.



when, $V_{in} < 2V$, $V_{out} = +V_{sat}$.
($V_- < V_+$)

when, $V_{in} > 2V$, $V_{out} = -V_{sat}$.
($V_- > V_+$)

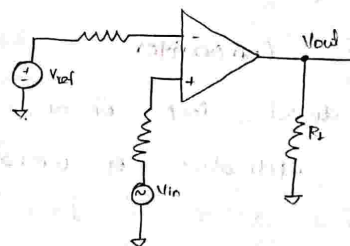
Assume, $V_{ref} = -2V$.



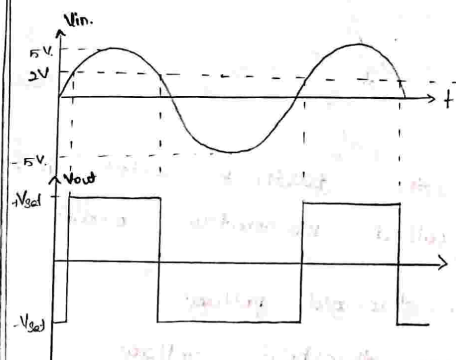
when, $V_{in} > -2V$, $V_o = -V_{sat}$.

$V_{in} < -2V$, $V_o = +V_{sat}$

Non-Inverting Mode comparator.



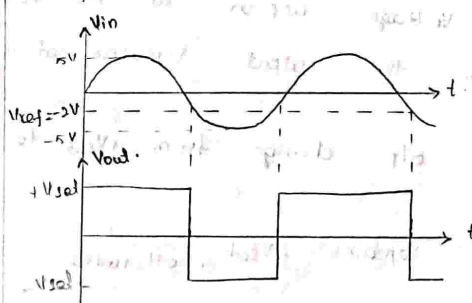
Assume, $V_{ref} = 2V$.



$V_{in} < 2V \Rightarrow V_{out} = -V_{sat}$
 $V_- < V_+$

$V_{in} > 2V \Rightarrow V_{out} = +V_{sat}$
 $V_- > V_+$

Assume, $V_{ref} = -2V$.



$V_{in} > -2V \Rightarrow V_o = +V_{sat}$

$V_{in} < -2V \Rightarrow V_o = -V_{sat}$
 $V_- < V_+$

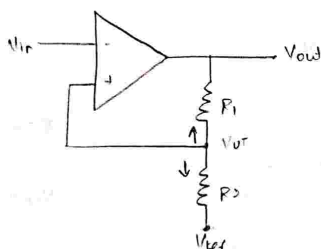
If we kept, $V_{sat} = 0V$ then it is zero crossing integrator circuit.

Bistable Regenerative Comparator:

⇒ Using closed loop op-amp.

⇒ It is application of comparator.

(also called as schmitt trigger)



whenever the feedback exist in the circuit is called regenerative circuit.

V_{UT} → upper threshold voltage.

V_{LT} → lower threshold voltage.

V_{UT} ⇒ Max Voltage appear at Non-inverting terminal
(upto which the output remains at $+V_{sat}$)

(or)
(when the o/p change from $+V_{sat}$ to $-V_{sat}$)

V_{LT} ⇒ output remains $-V_{sat}$, otherwise
It then

Minimum voltage, It remains $-V_{sat}$, otherwise the o/p transition from $-V_{sat}$ to $+V_{sat}$.

[Draw back of using open loop op-amp:

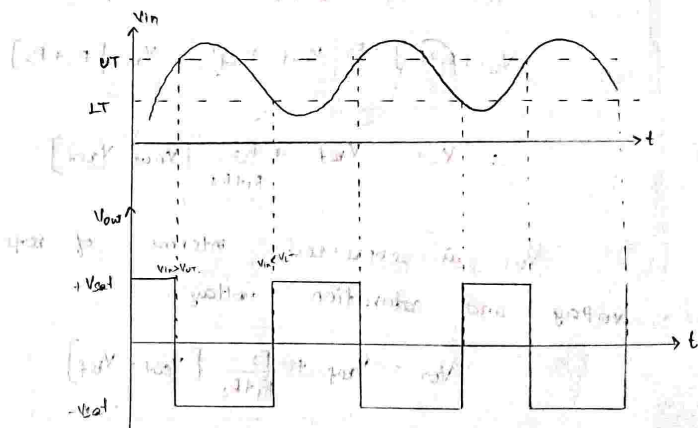
⇒ it has one threshold voltage]

⇒ o/p depends upon multiple threshold voltage [here we use 2].

when the i/p lies b/w V_{UT} & V_{LT} the o/p simply maintains its previous value. → hysteresis value.

V_{UT} → is a point max voltage at which the o/p change from $+V_{sat}$ to $-V_{sat}$.

V_{LT} → Max voltage at which o/p changes from $-V_{sat}$ to $+V_{sat}$.



when, $V_{out} = +V_{sat}$, $V_+ = V_{UT}$

apply nodal analysis, w.r.t. " V_{UT} "

$$\frac{V_{UT} - V_{out}}{R_1} + \frac{V_{UT} - V_{ref}}{R_2} = 0$$

$$V_{UT} \left[\frac{1}{R_1} + \frac{1}{R_2} \right] - \frac{V_{out}}{R_1} - \frac{V_{ref}}{R_2} = 0$$

$$V_{UT} \left[\frac{R_1 + R_2}{R_1 R_2} \right] = \frac{V_{out}}{R_1} + \frac{V_{ref}}{R_2}$$

$$V_{UT} [R_1 + R_2] = R_1 R_2 \left[\frac{V_{out}}{R_1} + \frac{V_{ref}}{R_2} \right]$$

$$V_{UT} [R_1 + R_2] = R_2 V_{out} + R_1 V_{ref}$$

add & sub by $R_2 V_{ref}$

$$V_{UT} [R_1 + R_2] = R_2 V_{out} + R_1 V_{ref} + R_2 V_{ref} - R_2 V_{ref}$$

$$V_{UT} [R_1 + R_2] = \frac{R_2 [V_{out} - V_{ref}]}{R_1 + R_2} + \frac{V_{ref} [R_1 + R_2]}{R_1 + R_2}$$

$$\therefore V_{UT} = V_{ref} + \frac{R_2}{R_1 + R_2} [V_{out} - V_{ref}]$$

V_{UT} is expressed in terms of reference voltage and saturation voltage.

$$V_{UT} = V_{ref} + \frac{R_2}{R_1 + R_2} [V_{sat} - V_{ref}]$$

when, $V_{out} = -V_{sat}$, $V_+ = V_{LT}$

$$\frac{V_{out} - V_{LT}}{R_1} + \frac{V_{ref} - V_{LT}}{R_2} = 0$$

apply nodal analysis w.r.t. V_{LT}

$$\frac{V_{LT} - V_{out}}{R_1} + \frac{V_{LT} - V_{ref}}{R_2} = 0$$

$$\frac{V_{LT} + V_{sat}}{R_1} + \frac{V_{LT} - V_{ref}}{R_2} = 0$$

$$V_{LT} \left[\frac{R_1 + R_2}{R_1 R_2} \right] = \frac{-V_{sat}}{R_1} + \frac{V_{ref}}{R_2}$$

xy by $R_1 R_2$:

$$V_{LT} [R_1 + R_2] = -R_2 V_{sat} + R_1 V_{ref} + R_2 V_{ref} - R_2 V_{ref}$$

$$V_{LT} [R_1 + R_2] = -R_2 [V_{sat} + V_{ref}] + V_{ref} [R_1 + R_2]$$

$$V_{LT} = \frac{-R_2}{R_1 + R_2} [V_{sat} + V_{ref}] + \frac{V_{ref} [R_1 + R_2]}{R_1 + R_2}$$

$$\therefore V_{LT} = V_{ref} - \frac{R_2}{R_1 + R_2} [V_{sat} + V_{ref}]$$

Hysteresis Voltage:

$$V_H = V_{UT} - V_{LT}$$

$$= \cancel{V_{ref}} + \frac{R_2}{R_1 + R_2} [V_{sat} - V_{ref}] - \cancel{V_{ref}} + \frac{R_2}{R_1 + R_2} (V_{sat} + V_{ref})$$

$$= \frac{R_2}{R_1 + R_2} [V_{sat} - V_{ref} + V_{sat} + V_{ref}]$$

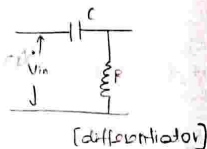
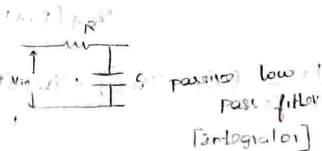
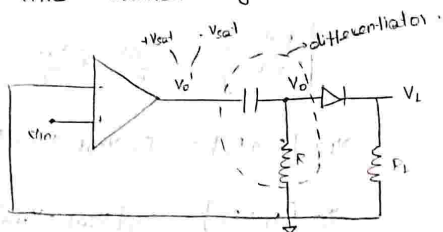
$$V_H = \frac{2 R_2}{R_1 + R_2} (V_{sat})$$

Advantage:

1. Noise immunity offered by regenerative comparator is high.
2. Switch speed is more.
3. Stability is more.

Application:

1. Time marker generative circuit.



$V_o = +Vsat$ $D \rightarrow 0$
 $V_o = -Vsat$ $D \rightarrow 1$

MULT-VIBRATORS:

Types of multivibrators.

1. Astable $\rightarrow$ No stable states / both are quasi stable states.
2. Monostable $\rightarrow$ 1 stable state & 1 quasi stable state.
3. Bistable $\rightarrow$ 2 stable states.

Astable $\Rightarrow$ no need to apply external i/p pulse to change its o/p states.

eg: oscillator, Relaxation oscillator.

- also called Free-Running multivibrator.

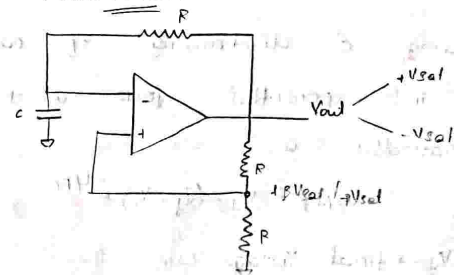
Monostable $\Rightarrow$ it requires -ve trigger pulse to change o/p from stable to quasi-stable state.

- also called one-shot multivibrator.

Bistable $\Rightarrow$ It requires two trigger pulse

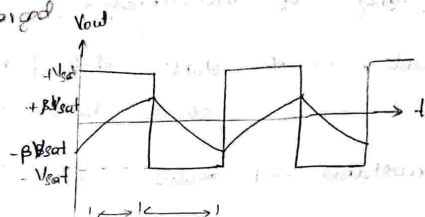
- also called binary circuit bcoz it has 2 stable state.

Astable Multivibrator!



→ Based on O/P

& discharged



→ o/p voltage depends upon threshold

voltage & charging/discharging of capacitor.

→ Initially Assume, $V_0 = +V_{sat}$, voltage at the terminal $+βV_{sat}$.

→ capacitor → charged beyond $β$ voltage

change $V_i > V_1$ then output is $-V_{sat}$

Assume, $V_0 = -V_{sat}$

$$V_1 = -βV_{sat}$$

C → discharge upto $-βV_{sat}$ when

it goes lower than $-βV_{sat}$

$V_i < V_1$ then the o/p is $+V_{sat}$

charging & discharging of capacitor

occur in exponential form, which is

mathematically as -

$$V_c(t) = V_f - (V_f - V_i) e^{-t/RC}$$

V_f → final voltage. (Max voltage 'c' can be charged is V_{sat}).

V_i → initial voltage.

T_1 - time period when capacitor charged exponentially

T_2 - time period when capacitor discharged exponentially.

By Mathematical,

$$V_c(t) = V_{sat} - (V_{sat} - (-βV_{sat})) e^{-t/RC}$$

$$V_c(t) = V_{sat} - (V_{sat} + βV_{sat}) e^{-t/RC}$$

at $t = T_1$, $V_c(t) = +βV_{sat}$.

$$+βV_{sat} = V_{sat} - (V_{sat} + βV_{sat}) e^{-T_1/RC}$$

$$βV_{sat} - V_{sat} = -(V_{sat} + βV_{sat}) e^{-T_1/RC}$$

$$+V_{sat}(1 - β) = +V_{sat}(1 + β) e^{-T_1/RC}$$

$$\frac{1 - β}{1 + β} = e^{-T_1/RC}$$

Apply ln on BLS,

$$-T_1/RC = \ln\left(\frac{1 - β}{1 + β}\right)$$

$$\frac{T_1}{RC} = -\ln\left(\frac{1 - β}{1 + β}\right)$$

$$\therefore T_1 = RC \ln\left(\frac{1 + β}{1 - β}\right)$$

capacitor required same time for

charging & discharging.

If o/p is symmetrical,

$$T = T_1 + T_2 = 2T_1$$

$$T = 2RC \ln \left(\frac{1+\beta}{1-\beta} \right)$$

$\therefore$ Time period depends on R, C & β .

If we don't want V_{sat} in o/p we can use clipping circuit by zener diode.

connected (1 by 1 zener diode) in V_{out} , it limits the amplitude.

$$\beta = \frac{R_2}{R_1 + R_2}$$

$$\text{If } R_1 = R_2,$$

$$\beta = \frac{R_1}{R_1 + R_1} = \frac{1}{2} = 0.5$$

$$T = 2RC \ln \left[\frac{1+0.5}{1-0.5} \right] \Rightarrow T = 2RC \ln [3]$$

$$\text{If } R_1 = 1.16 R_2,$$

$$\beta = \frac{R_1}{1.16 R_2} = \frac{1}{1.16} = 0.463$$

$$T = 2RC \ln \left[\frac{1+0.463}{1-0.463} \right] = 2RC \ln [2.724]$$

$$T = 2RC (1.002)$$

$$T_1 = 2RC$$

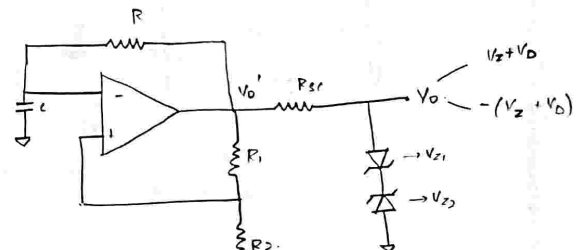
In this case time period is completely depends on external R, C .

$$\therefore f_0 = \frac{1}{T} = \frac{1}{2RC}$$

It is a symmetrical square wave generator circuit.

Symmetrical - on & off time is same.

For asymmetry square wave:



I am going to fix V_0 value by taking zener diode back to back. Standard voltage of zener diode $\approx 5.6V$.

$R_{sc} \rightarrow$ current limiting resistor

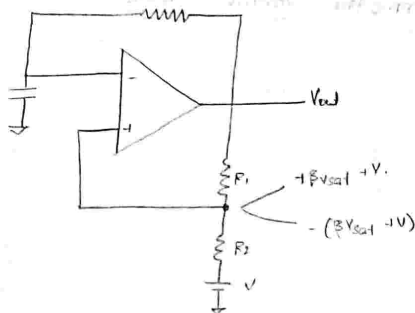
(in order to prevent the breakdown of zener by allowing excessive current in $R.B$).

By connecting zener diode
able to fix voltage $(V_Z + V_D) / -(V_Z + V_D)$
rather than $+V_{sat}, -V_{sat}$.

$$I = \frac{V_{out} - V_Z}{R_{sc}} \quad V_Z \rightarrow \text{breakdown voltage}$$

$V_{Z1} = V_{Z2} \Rightarrow$ generate symmetric square wave.

$V_{Z1} \neq V_{Z2} \Rightarrow$ generate asymmetric " " " " " "
asymmetric square wave generator.



[application of Astable:

* Voltage to frequency converter.]

$\Rightarrow$ We can control the frequency
by changing the dc voltage.

$\Rightarrow$ The o/p of astable depends on
external dc supply.

Design a square wave oscillator for
 $f_o = 1 \text{ kHz}$ using an JCT41 op-amp and dc
supply voltage of $\pm 10 \text{ V}$

sol: Given, $f_o = 1 \text{ kHz}$.

Design a symmetric square wave.

$$f_o = \frac{1}{2RC}$$

$$R_1 = 1.16 R_2$$

$$\text{Let } R_2 = 10 \text{ k}\Omega$$

$$R_1 = 11.6 \text{ k}\Omega$$

$$f_o = \frac{1}{2RC} = 1 \text{ kHz}$$

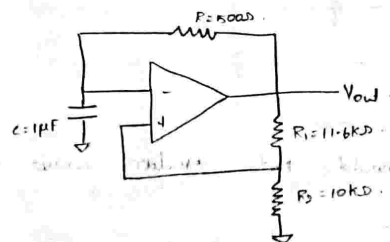
let us choose, $C = 1 \mu\text{F}$.

$$\frac{1}{2 \times 1 \times 10^{-6} \times 10^3} = R$$

$$\frac{1}{2 \times 10^{-3}} = R$$

$$R = 0.5 \text{ k}\Omega$$

$$R = 500 \Omega$$



MONOSTABLE MULTIVIBRATOR:

1). It has 1 stable state & 1 quasi stable state.

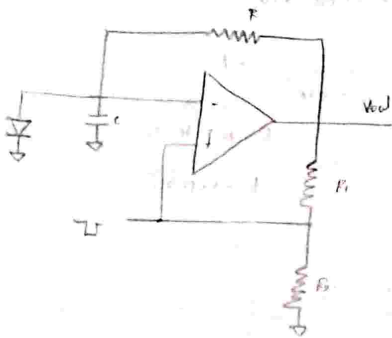
$\Rightarrow$ quasi stable state $\rightarrow$ Non-permanent state.

2). It requires 1 triggering pulse externally. (to change state of op)

3). It is called 1-shot multi-vibrator.

4). Gating circuits - It can generate a pulse (or) square wave in specified time period (fixed time period).

5). Time delay circuit.



It should not produce wave form continuously.

By placing the diode in parallel to capacitor, when 0.7V the capacitor stop charging (so that it doesn't produce continuous).

$\rightarrow V_{out} = +V_{sat}$. (Before applying -ve trigger)

The voltage at capacitor fixed at ' V_0 '

Now, I need to apply with suitable amplitude. (giving -ve max voltage.)

eg: $-4.00V < 0.00V$
 $\downarrow$ $\downarrow$
 -ve trigger pulse. inverting terminal voltage.

[trigger $\rightarrow$ short duration pulse].

$\therefore$ Inverting terminal $>$ Non-inverting voltage, the op $\rightarrow -V_{sat}$.

$\Rightarrow$ Now, capacitor start discharge upto $-V_{sat}$.

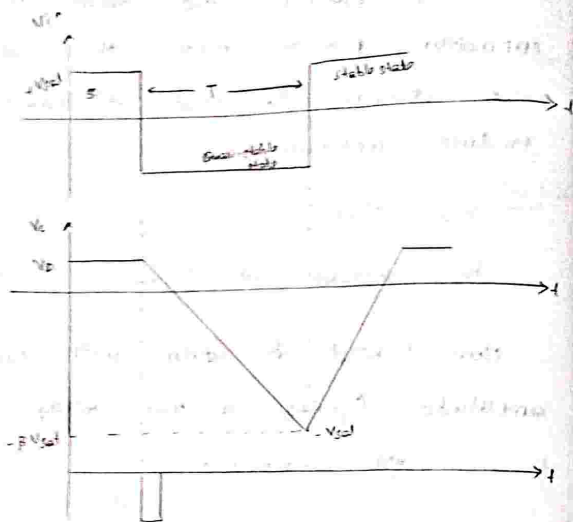
$\Rightarrow$ when it goes below $-V_{sat}$,

$V_- < V_+$ (-ve trigger) like $(-5.00 < -4.00)$,
 V_- V_+ (-ve trigger)

o/p $\rightarrow +V_{sat}$

$\Rightarrow$ due to in parallel connection of diode, capacitor discharge linearly.

after it charge upto $+V_{sat}$, but due to diode charge upto 0.7V after that remains V_0 .



$$V_c(t) = V_i - (V_i - V_i) e^{-t/RC}$$

$$V_c(t) = -V_{sat} - (-V_{sat} - V_D) e^{-t/RC}$$

$$\Rightarrow \therefore V_c(t) = -\beta V_{sat}, \text{ at } t = T$$

$$-\beta V_{sat} = -V_{sat} - (-V_{sat} - V_D) e^{-T/RC}$$

$$-\beta V_{sat} + V_{sat} = [V_{sat} + V_D] e^{-T/RC}$$

$$\frac{V_{sat} [1 - \beta]}{V_{sat} + V_D} = e^{-T/RC}$$

$$e^{T/RC} = \left[\frac{1 - \beta}{1 + \frac{V_D}{V_{sat}}} \right]$$

$$e^{T/RC} = \frac{1 + \frac{V_D}{V_{sat}}}{1 - \beta}$$

take ln on b/s,

$$\frac{T}{RC} = \ln \left(\frac{1 + \frac{V_D}{V_{sat}}}{1 - \beta} \right)$$

$$T = RC \ln \left(\frac{1 + \frac{V_D}{V_{sat}}}{1 - \beta} \right)$$

$\therefore V_D \rightarrow$ diode drop. (less $\approx 0.7V$).

$V_{sat} \rightarrow$ very high

if $V_D < V_{sat}$, $\frac{V_D}{V_{sat}} \ll 1$, so it is neglected.

$$T = RC \ln \left[\frac{1}{1 - \beta} \right], \text{ here } \beta = \frac{R_2}{R_1 + R_2}$$

if $R_1 = R_2$,

$$\beta = \frac{R_2}{R_1 + R_2} = 0.5$$

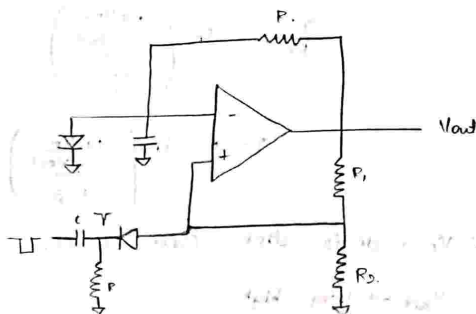
$$T = RC \ln \left[\frac{1}{1 - 0.5} \right] = RC \ln \left[\frac{1}{0.5} \right]$$

$$T = RC \ln(2)$$

$$\therefore T = 0.693 RC$$

By choosing RC, we can determine generate a pulse of T.

diode is connected to ensure the
-no trigger is applied, when the
trigger apply diode become OFF.



- ⑤. Design a monostable multivibrator with trigger pulse shaping which will drive a LED ON for 0.5 sec, each time it is pulsed.

Sol: Given, $T = 0.5 \text{ sec.}$

Formula, $T = 0.69 RC$

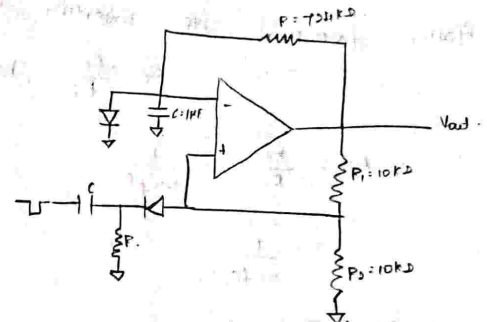
Let us choose, $C = 1 \mu F$

$$R = \frac{0.5}{0.69 \times 1 \times 10^{-6}}$$

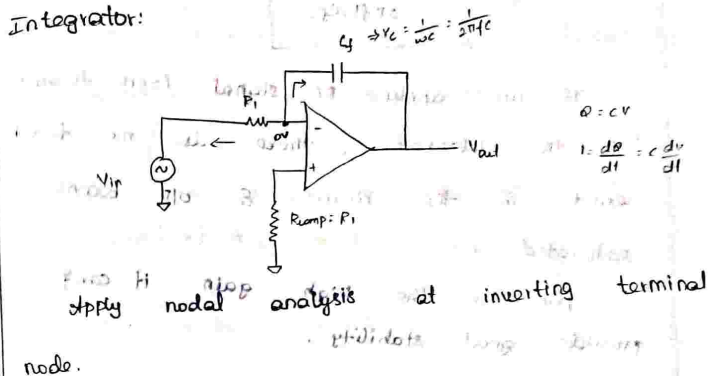
$$R = 0.7246 \times 10^6$$

$$R = 724.63 \text{ K}\Omega$$

assume, $R_1 = R_2 = 10 \text{ K}\Omega$



Integrator:



$$\frac{0 - V_{in}}{R_1} + \frac{0 - V_{out}}{R_3} = 0$$

$$-\frac{C dV_{out}}{dt} = \frac{V_{in}}{R_1}$$

$$\frac{dV_{out}}{dt} = -\frac{V_{in}}{R_1 C_f}$$

$$\therefore V_{out} = -\frac{1}{R_1 C_f} \int V_{in} \cdot dt + V_c(0)$$

$V_c(0) \rightarrow$ initial voltage at capacitor

Gain, [compare with inverting amplifier]

$$A_c = -\frac{R_f}{R_i}, \text{ here } R_f = X_c$$

$$\therefore A_c = -\frac{X_c}{R_i} = -\frac{1}{\omega C_f R_i}$$

$$= -\frac{1}{2\pi f C_f R_i}$$

$$|A_c| = \frac{1}{2\pi f R_i C_f}$$

If we apply DC signal (zero frequency) to the integrator, there is no feedback exist in the circuit, & o/p becomes saturated. (bec it becomes open loop)

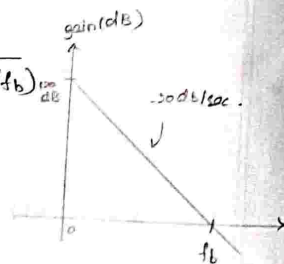
Due to the high gain it can't provide good stability.

open loop gain practical $\approx 10^6$

$$\begin{aligned} \text{gain in dB} &= 20 \log_{10}(10^6) \\ &= 20 \times 6 \\ &= 120 \text{ dB} \end{aligned}$$

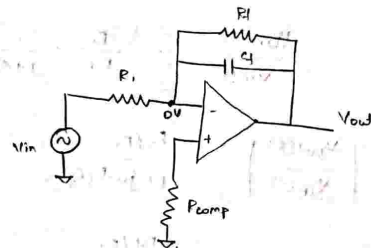
$$|A_c| = \frac{1}{2\pi f R_i C_f}$$

$$\therefore f_b = \frac{1}{2\pi R_i C_f}$$



cutoff frequency at which gain of circuit drops to 0 dB.

Practical Integrator: (lossy integrator). To overcome the drawback we add one more resistor in feedback.



Now, if we give 0 Hz as frequency, the impedance is ∞ it becomes open circuit, now it is similar as inverting amplifier. so o/p doesn't saturate. Now the gain is $-\frac{R_f}{R_i}$

Apply Nodal Analysis.

$$\frac{0 - V_{in}}{R_i} + \frac{0 - V_{out}}{R_f} - C \frac{dV_{out}}{dt} = 0$$

$$-C \frac{dV_{out}}{dt} = \frac{V_{in}}{R_i} + \frac{V_{out}}{R_f}$$

take Laplace transform.

$$-s C_f V_{out}(s) = \frac{V_{in}(s)}{R_i} + \frac{V_{out}(s)}{R_f}$$

$$\frac{V_{out}(s)}{R_f} + s C_f V_{out}(s) = -\frac{V_{in}(s)}{R_i}$$

$$V_{out}(s) \left[\frac{1}{R_f} + s C_f \right] = -\frac{1}{R_i} V_{in}(s)$$

$$V_{out}(s) \left[\frac{1+sR_fC_f}{R_f} \right] = -\frac{1}{R_i} V_{in}(s)$$

$$\frac{V_{out}(s)}{V_{in}(s)} = -\frac{R_f}{R_i} \cdot \frac{1}{1+sR_fC_f}$$

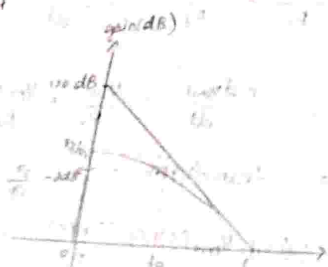
$$\left| \frac{V_{out}(s)}{V_{in}(s)} \right| = \frac{R_f/R_i}{1+j\omega R_fC_f}$$

$$= \frac{R_f/R_i}{1+j\omega R_fC_f}$$

$$\left| \frac{V_{out}(s)}{V_{in}(s)} \right| = \frac{R_f/R_i}{1+j\omega R_fC_f}$$

$$\therefore f_0 = \frac{1}{2\pi R_f C_f}$$

At $f=0$, gain is $\frac{R_f}{R_i}$ which is voltage gain.
When $f=f_0$, the gain is reduced to -3 dB .



when we design practical integrator we need to follow three rules:

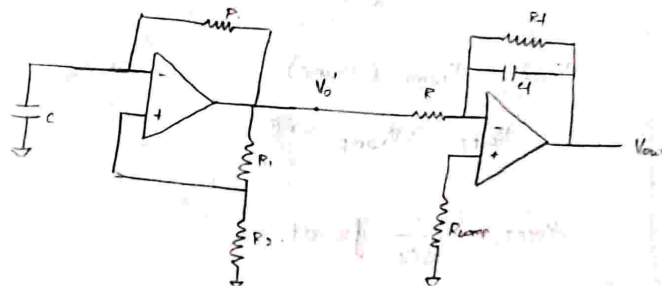
- ① Relationship between R_f & R_i , $\Rightarrow R_f = 10R_i$
- ② $f_{in} = 10 \cdot f_0$
- ③ $f_{oc} \leq f_L$

$\Rightarrow$ without R_f , C_f charge completely full

$\Rightarrow$ with R_f , amount of charging by capacitor is reduced, no possible for full (or) complete charge.

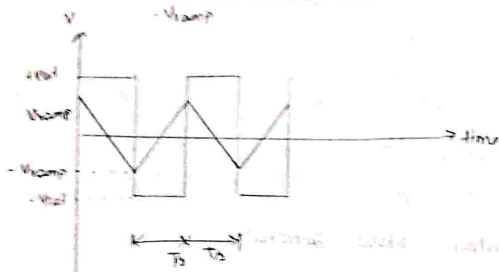
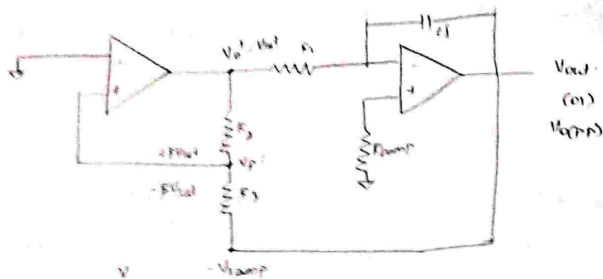
current divide & pass through R_f & C_f so leakage of current occurs. so it is called lossy (loss of power due to R_f)

Triangular Wave generator:



In order to determine the frequency of square wave generator, remove R & C .

In order to change voltage of square of wave generator, connect output with R_f if.



$$V_o(pp) = V_{ramp} - (-V_{ramp})$$

$$V_o(pp) = 2V_{ramp} \rightarrow \text{---} \text{---}$$

$$V_o(pp) = \frac{1}{R_1 C_f} \int V_o' dt$$

Initially use equation, $V_o' = +V_{sat}$

$$V_{out} = -V_{ramp}$$

Applying nodal analysis with respect to V_p

$$\frac{V_p - V_{sat}}{R_1} + \frac{V_p - (-V_{ramp})}{R_2} = 0$$

$$V_p \left[\frac{1}{R_1} + \frac{1}{R_2} \right] = \frac{V_{sat}}{R_1} - \frac{V_{ramp}}{R_2}$$

$$V_p \left(\frac{R_2 + R_1}{R_1 R_2} \right) = \frac{V_{sat}}{R_1} - \frac{V_{ramp}}{R_2}$$

$$V_p (R_2 + R_1) = R_2 V_{sat} - R_2 V_{ramp}$$

$$V_p (R_2 + R_1) = R_2 V_{sat} - R_2 V_{ramp} + R_2 V_{ramp} - R_2 V_{ramp}$$

$$V_p (R_2 + R_1) = R_2 (V_{sat} + V_{ramp}) - V_{ramp} (R_2 + R_1)$$

$$V_p = \frac{R_2}{R_2 + R_1} (V_{sat} + V_{ramp}) - V_{ramp}$$

$V_p \rightarrow$ is point at which the transition from $+V_{sat}$ to $-V_{sat}$

$$\text{make, } \therefore V_p = 0$$

$$\frac{R_2}{R_2 + R_1} [V_{sat}] + \frac{R_2}{R_2 + R_1} V_{ramp} - V_{ramp} = 0$$

$$\frac{R_2}{R_2 + R_1} V_{sat} + V_{ramp} \left[\frac{R_2}{R_2 + R_1} - 1 \right] = 0$$

$$V_{ramp} \left[\frac{R_2 - R_2 - R_1}{R_2 + R_1} \right] + \left(\frac{R_2}{R_2 + R_1} \right) V_{sat} = 0$$

$$R_2 V_{sat} = R_2 V_{ramp}$$

$$\therefore V_{ramp} = \frac{R_2}{R_1} V_{sat}$$

$$\therefore V_o(pp) = 2V_{ramp} = \frac{2R_2}{R_1} V_{sat} \quad \text{---} \text{---}$$

$$V_{out}(pp) = \frac{1}{R_1 C_f} \int_0^{T/2} V_o' dt$$

$$= -\frac{1}{R_1 C_f} \int_0^{T/2} (-V_{sat}) dt$$

$$= \frac{1}{R_1 C_f} V_{sat} \cdot [t]_0^{T/2}$$

$$V_{out(pp)} = \frac{T \cdot V_{sat}}{2 R_1 C_f} \rightarrow (3)$$

equate (2) & (3)

$$\frac{2 R_3}{R_1} V_{sat} = \frac{T V_{sat}}{2 R_1 C_f}$$

$$T = \frac{4 R_1 R_3 C_f}{R_2}$$

This is time period of triangular wave generator.

$$\therefore f = \frac{1}{T} = \frac{R_2}{4 R_1 R_3 C_f}$$

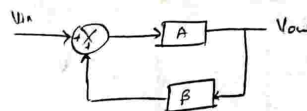
OSCILLATOR:

It is a electronic device, to produce a sinusoidal (or) non-sinusoidal signal periodically (continuously) without applying external input.

$$\text{-ve F.B} \rightarrow \frac{V_o(s)}{V_i(s)} = \frac{G}{1+GH} \quad (\text{or}) \quad \frac{A}{1+A\beta}$$

$$\text{+ve F.B} \rightarrow \frac{V_o(s)}{V_i(s)} = \frac{G}{1-GH} \quad (\text{or}) \quad \frac{A}{1-A\beta}$$

* crystal oscillator provide high stability.



Barkhausen criterion: $|AB| = 1$

$$\phi = 0^\circ \text{ or } 360^\circ$$

* If oscillator has -ve F.B, we won't get constant amplitude. (It can't generate continuously)

* To design a oscillator, the circuit should have +ve feedback. to get continuously $\rightarrow$ should be in-phase.

If $A\beta < 1$, $A\beta > 1$; we won't get continuous wave without apply external input.

RC oscillators:

The feedback network is designed only with the help (use) of resistor & capacitor.

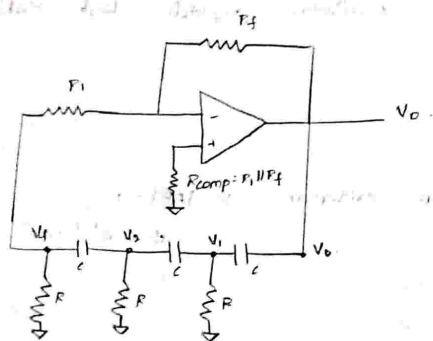
Drawback:

- * The freq of signal the frequency is in Hz (or) kHz.
- * stability is low

① R-C phase shift oscillator.

② Wien-bridge oscillator.

③ RC phase shift oscillator.

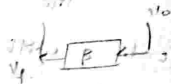


op-amp provide -180° phase shift
 feedback network - 180° phase shift
 provide $= 360^\circ$

In T-B network, each RC network provide 60° phase shift to satisfy, $|A\beta| = 1$.

$$\beta = \frac{V_f}{V_o} \quad [\text{feedback gain}]$$

$$\therefore V_f = \beta V_o$$



Sustained oscillation:

It generates waveform with

constant amplitude & frequency.

$$|A\beta| = 1$$

If $|A\beta| < 1$, the amplitude decreases.
 If $|A\beta| > 1$, the output becomes saturated.

write down KCL with respect to node 1 V_1 .

$$\frac{V_1 - V_o}{X_c} + \frac{V_1 - 0}{R} + \frac{V_1 - V_2}{X_c} = 0$$

$$X_c = \frac{1}{sC}$$

$$(V_1 - V_o)sc + \frac{V_1}{R} + (V_1 - V_2)sc = 0$$

$$V_1 sc + \frac{V_1}{R} + V_1 sc - V_o(sc) - V_2(sc) = 0$$

$$V_1 \left[2sc + \frac{1}{R} \right] - V_o(sc) - V_2(sc) = 0$$

$$V_1 \left[\frac{1 + 2sCR}{R} \right] - V_o(sc) - V_2(sc) = 0$$

$$V_1 (1 + 2sCR) - V_o(Rsc) - V_2(sCR) = 0$$

$$V_1 [1 + 2sCR] - V_2[sCR] = V_o[sCR]$$

$$V_o = \frac{V_1(1 + 2sCR)}{sCR} - V_2 \rightarrow \text{①}$$

write down KCL with respect to

node 2 V_2

$$\frac{V_2 - V_1}{X_c} + \frac{V_2 - 0}{R} + \frac{V_2 - V_3}{X_c} = 0$$

$$(V_2 - V_1)(sc) + \frac{V_2}{R} + (V_2 - V_3)(sc) = 0$$

$$V_2 sC + \frac{V_2}{R} + V_2 sC - V_1(sC) - V_f(sC) = 0$$

$$V_2 \left[2sC + \frac{1}{R} \right] - V_1(sC) - V_f(sC) = 0$$

$$V_2 \left[\frac{1+2sCR}{R} \right] - V_1(sCR) - V_f(sCR) = 0$$

$$V_f(sCR) = V_2(1+2sCR) - V_1(sCR)$$

$$V_f = \frac{V_2(1+2sCR)}{sCR} - V_1 \rightarrow \textcircled{D}$$

write down KCL with respect to node 2, V_f

$$\frac{V_f - V_2}{R} + \frac{V_f - 0}{R} = 0$$

$$(V_f - V_2)sC + \frac{V_f}{R} = 0$$

$$V_f \left[sC + \frac{1}{R} \right] - V_2(sC) = 0$$

$$V_f \left[\frac{1+RSC}{R} \right] - V_2(sC) = 0$$

$$V_f(1+RSC) - V_2(sCR) = 0$$

$$V_2 = \frac{V_f(1+RSC)}{sCR} \rightarrow \textcircled{B}$$

sub $\textcircled{B}$ in $\textcircled{D}$

$$V_f = \frac{V_f(1+RSC)}{sCR} \cdot \frac{(1+2sCR)}{sCR} - V_1$$

$$V_f = \frac{V_f(1+RSC)(1+2sCR)}{s^2CR^2} - V_1$$

$$V_f = \frac{V_f(1+2sCR + sCR + 2s^2R^2C^2)}{s^2R^2C^2} - V_1$$

$$-V_f + \frac{V_f(1+3sCR + 2(sCR)^2)}{(sCR)^2} = V_1$$

$$\frac{-V_f(sCR)^2 + V_f(1+3sCR + 2(sCR)^2)}{(sCR)^2} = V_1$$

$$\frac{V_f[-(sCR)^2 + 1 + 3sCR + 2(sCR)^2]}{(sCR)^2} = V_1$$

$$V_1 = \frac{V_f[(sCR)^2 + 1 + 3sCR]}{(sCR)^2}$$

$$V_1 = \frac{V_f[1 + 3sCR + (sCR)^2]}{(sCR)^2}$$

sub V_1, V_2 in $\textcircled{D}$

$$V_0 = \frac{V_f[1+3sCR + (sCR)^2]}{(sCR)^2} \cdot \frac{(1+2sCR)}{(sCR)} - \frac{V_f(1+RSC)}{sCR}$$

$$= \frac{V_f(1+3sCR + (sCR)^2)(1+2sCR)}{(sCR)^3} - \frac{V_f(1+RSC)}{sCR}$$

$$= \frac{V_f[1+3sCR + (sCR)^2](1+2sCR) - V_f(1+RSC)(sCR)^2}{(sCR)^3}$$

$$= \frac{V_f[1+3sCR + 3sCR + 6(sCR)^2 + (sCR)^2 + 2(sCR)^3] - V_f[(sCR)^2 + (sCR)^3]}{(sCR)^3}$$

$$V_o = \frac{V_f [1 + 5sRC + 7(sRC)^2 + 2(sRC)^3] - V_f [(sRC)^2 + (sRC)^3]}{(sRC)^3}$$

$$V_o = \frac{V_f [1 + 5sRC + 7(sRC)^2 + 2(sRC)^3 - (sRC)^2 - (sRC)^3]}{(sRC)^3}$$

$$V_o = \frac{V_f [1 + 5sRC + 6(sRC)^2 + (sRC)^3]}{(sRC)^3}$$

$$\frac{V_f}{V_o} = \frac{(sRC)^3}{[1 + 5sRC + 6(sRC)^2 + (sRC)^3]}$$

$$\beta = \frac{(sRC)^3}{(sRC)^3 \left[\frac{1}{(sRC)^3} + 5 \frac{1}{(sRC)^2} + \frac{6}{sRC} + 1 \right]}$$

$$\beta = \frac{V_f}{V_o} = \left[\frac{1}{1 + \frac{5}{(sRC)^2} + \frac{6}{sRC} + \frac{1}{(sRC)^3}} \right]$$

Replace s with $j\omega$ and ω with

$2\pi f$.

$$\beta = \frac{1}{1 + \frac{6}{j2\pi fRC} + \frac{5}{4\pi^2 f^2 R^2 C^2} + \frac{1}{(2\pi fRC)^3}}$$

$$\beta = \frac{1}{\left[1 - 5 \left(\frac{f_1}{f} \right)^2 \right] + \frac{1}{5} \left[6 \left(\frac{f_1}{f} \right) - \left(\frac{f_1}{f} \right)^3 \right]}$$

$$f_1 = \frac{1}{2\pi RC}$$

$$= \frac{1}{\left[1 - 5 \left(\frac{f_1}{f} \right)^2 \right] - j \left[6 \left(\frac{f_1}{f} \right) - \left(\frac{f_1}{f} \right)^3 \right]}$$

The imaginary term is zero.

$$\beta = \frac{1}{\left(1 - 5 \left(\frac{f_1}{f} \right)^2 \right)} \rightarrow (3)$$

$$\left[6 \left(\frac{f_1}{f} \right) - \left(\frac{f_1}{f} \right)^3 \right] = 0$$

$$6 \left(\frac{f_1}{f} \right) = \left(\frac{f_1}{f} \right)^3$$

$$\frac{f_1}{f} = \sqrt{6} \rightarrow (4)$$

$$f_0 = \frac{1}{\sqrt{6} 2\pi RC}$$

Sub (11) in (3),

$$\beta = \frac{1}{1 - 5(6)}$$

$$= \frac{1}{1 - 30} = \frac{1}{-29}$$

$$A_v \cdot \beta = 1$$

$$A_v \cdot \left(-\frac{1}{29} \right) = 1$$

$$A_v \geq -29$$

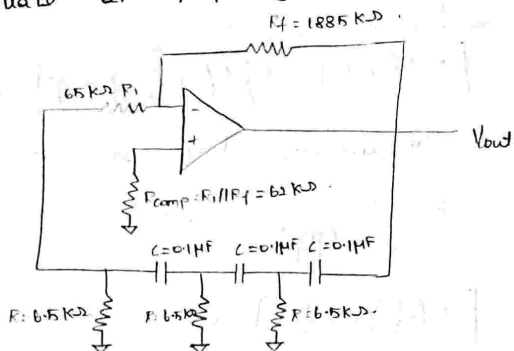
To find R_1 & R_f .

$$1 - \frac{R_f}{R_1} = -29$$

$$R_f = 29 R_1$$

$$R_1 = 10 R$$

6). Design an RC phase shift oscillator to oscillate at frequency $f_0 = 100\text{Hz}$.



$$f_0 = \frac{1}{\sqrt{6} \times \pi R C}$$

Let $C = 0.1 \mu\text{F}$.

$$100 = \frac{1}{\sqrt{6} \times 2 \times 3.14 \times R \times 0.1 \times 10^{-6}}$$

$$100 = \frac{1}{1.5385 \times 10^{-6} \times R}$$

$$\frac{1}{R} = 153.85 \times 10^{-6}$$

$$R = 6.501 \times 10^6 \times 10^{-3}$$

$$R = 6.5 \text{ k}\Omega$$

$$R_f = 10(R)$$

$$= 10 \times 6.5 \times 10^3$$

$$R_f = 65 \text{ k}\Omega$$

$$R_f = 29(65 \text{ k}\Omega)$$

$$R_f = 1885 \text{ k}\Omega$$

$$\frac{1}{R_{\text{comp}}} = \frac{1}{R_f} + \frac{1}{R_f}$$

$$\frac{1}{R_{\text{comp}}} = \frac{1}{65000} + \frac{1}{1885000}$$

$$\frac{1}{R_{\text{comp}}} = \frac{1885000 + 65000}{(65000)(1885000)}$$

$$= \frac{1950000}{1.225 \times 10^{11}}$$

$$= \frac{1950 \times 10^3}{1.225 \times 10^{11}}$$

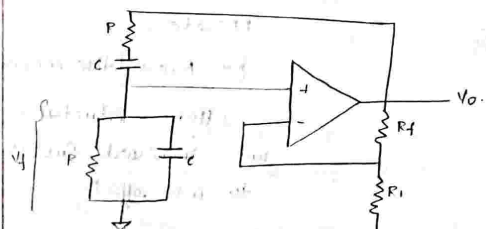
$$\frac{1}{R_{\text{comp}}} = 1591.83 \times 10^{-8}$$

$$R_{\text{comp}} = 6.283 \times 10^{-4} \times 10^8$$

$$= 6.283 \times 10^4$$

$$R_{\text{comp}} = 62 \text{ k}\Omega$$

WIEEN - BRIDGE OSCILLATOR:



Active Filters:

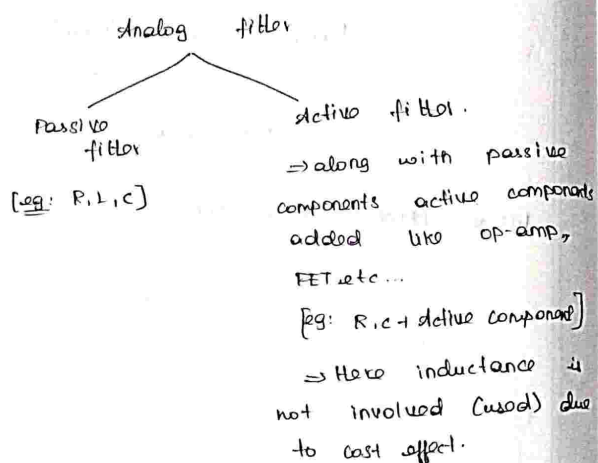
Filter $\rightarrow$ Frequency selective Network.

* Based on nature of signals as i/p.

Filter is classified in.

1. Analog filter.
2. Digital filter.

* In analog filter is classified based on the components used.



* Active filter based on suppression of i/p signal is classified as,

1. Low pass filter - LPF
2. High pass filter - HPF
3. Band pass filter - BPF.

1). BPF / BRF / BPF

* Based on frequency response, the filter classification.

reduction in gain occurs in monotonically also.
1. Butterworth filter - flat PB, flat SB.

[You may not find the ripple]

2. Chebyshev filter

3. Bessel function.

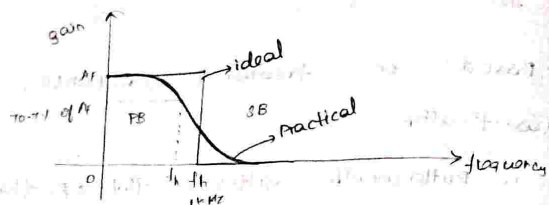
$\Rightarrow$ We use Butterworth filters.

Parameters:

- 1). Pass Band.
- 2). Stop Band.
- 3). Cut-off frequency.

Let us consider low pass filter.
It $\rightarrow$ maximum range of freq. that is transmitted by LP filter.

stop band $\rightarrow$ Range of frequency of i/p signal that can be suppressed/eliminated/attenuated by filter circuit.



For ideal characteristics of LPF,

PB = 100% gain. [we get exact transition]
SB = No gain.

For Practical,

* We can't get exact transition from high to low.

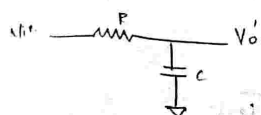
* It is monotonically decreasing (gain) as frequency is.

f_h → frequency at which the separation (of) transition from PB to SB. [ideal].
[cutoff / corner freq]

f_h → It is frequency at which the gain reduced by 3 dB.

The gain which is reduced to 10.7% of its maximum gain, that frequency is cut-off frequency [Practical].

First order butlerworth active LPF:



$$V_o' = \frac{V_{in}}{R + 1/sC} = \frac{1}{sC} V_{in}$$

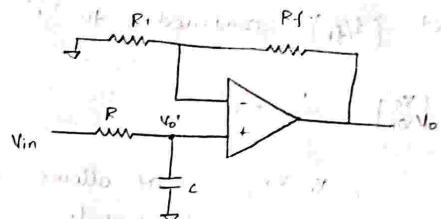
Passive LPF

→ max gain is, $A = 1$.

where ($\omega = 0$)

$$A = \frac{V_o'}{V_{in}} = \frac{1}{1 + j\omega RC}$$

→ when $\omega = \infty$, $A = 0$.



$$V_o = \left(1 + \frac{R_f}{R_i}\right) V_o' \rightarrow \text{①}$$

Advantage:

→ In order to improve the gain of passive LPF we added active component. that advantage of active component is applied to active filter.

→ By choosing proper R_f & R_i

we can improve the gain.

$$V_o = A_F \cdot \frac{V_{in}}{1 + j\omega RC}$$

Advantage: 2nd

1. No loading effect

in active filter

→ small size components

to passive filter.

$$= \frac{A_F}{1 + j\omega RC} \cdot V_{in}$$

→ low cost because

[inductance is not involved].

$$\frac{V_o}{V_{in}} = \frac{A_F}{1 + j(f/f_h)} \rightarrow \text{②}$$

where, $f_h = \frac{1}{2\pi RC}$.

$$\left| \frac{V_o}{V_{in}} \right| = \frac{1}{\sqrt{1 + \left(\frac{f}{f_h} \right)^2}}$$

case (i): $f \ll f_h$.

Neglect $\left(\frac{f}{f_h} \right)$ compared to '1'

$$\left| \frac{V_o}{V_{in}} \right| = \frac{1}{1} = 1.$$

$V_o = V_{in}$. $\therefore$ It allows all the i/p signals.

case (ii): $f \gg f_h$.

neglect 1, $1 + \left(\frac{f}{f_h} \right)^2 \approx \left(\frac{f}{f_h} \right)^2$

$$\left| \frac{V_o}{V_{in}} \right| = \frac{1}{\sqrt{\left(\frac{f}{f_h} \right)^2}} = \frac{1}{f/f_h}$$

$$\left| \frac{V_o}{V_{in}} \right| = \frac{f_h}{f}$$

$\therefore$ signal is get attenuated (decreased).

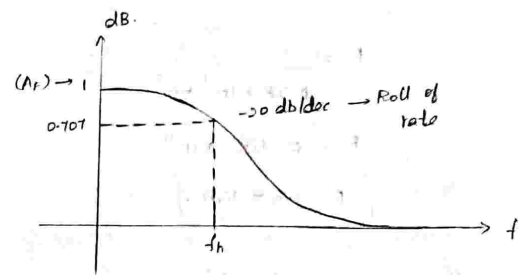
$\therefore$ It suppress the all i/p signal.

case (iii): $f = f_h$.

$$\left| \frac{V_o}{V_{in}} \right| = \frac{1}{\sqrt{1 + (1)^2}} = \frac{1}{\sqrt{2}}$$

$$\left| \frac{V_o}{V_{in}} \right| = 0.707$$

The gain of circuit is reduced to 0.707 times of maximum gain.



In General,

Roll-off rate of n th order filter is.

-n 20 dB/dec.



7. Design a Butterworth with a cutoff frequency of 1 kHz.

Sol:
$$f_h = \frac{1}{2\pi RC} = 1 \times 10^3 \text{ Hz.}$$

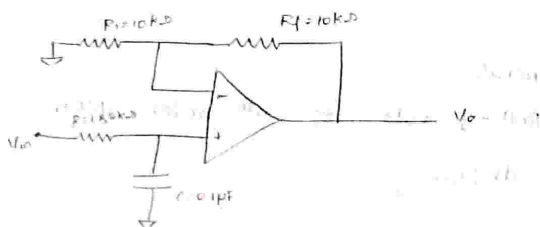
Let us take, $C = 0.1 \mu\text{F}$.

$$10^3 = \frac{1}{2 \times 3.14 \times R \times 0.1 \times 10^{-6}}$$

$$R = \frac{1}{6.28 \times 10^{-7} \times 10^3}$$

$$R = 0.159 \times 10^4$$

$$R = 1.59 \text{ K}\Omega$$



here, $1 + \frac{R_f}{R_i} = A_f = 1$.

assume $R_i = R_f = 10 \text{ K}\Omega$.

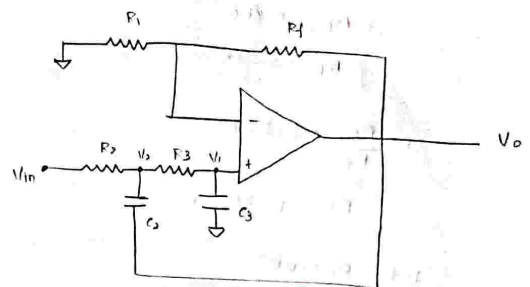
$$1 + \frac{R_f}{R_f} = A_f$$

$$1 + 1 = A_f$$

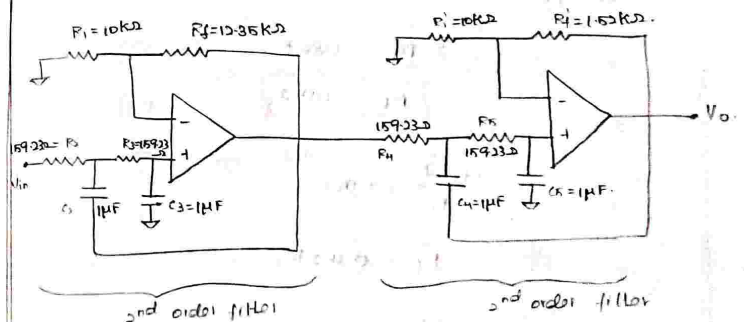
$$\text{gain, } A_f = 2$$

2nd order Butterworth LPF:

The order of Butterworth LPF which strongly no. of RC section connected in the circuit.



8. Design a Butterworth 4th order LPF with cutoff frequency of 1 kHz.



Sol:

Given, $f_h = 1 \text{ KHz}$.

W.K.T, $(s^2 + 0.765s + 1)(s^2 + 1.848s + 1)$

→ ①.

From 2nd order Butterworth LPF,

$$s^2 + \frac{(3-A_F)}{R_C} s + \frac{1}{R_C^2} \rightarrow \textcircled{2}$$

compare $\textcircled{1}$ & $\textcircled{2}$, for 1st & 2nd order.

$$3-A_F = 0.765$$

$$A_F = 2.235$$

$$1 + \frac{R_f}{R_1} = 2.235$$

$$R_f = 1.235 R_1$$

$$\text{let } R_1 = 10 \text{ k}\Omega$$

$$R_f = 12.35 \text{ k}\Omega$$

compare $\textcircled{1}$ & $\textcircled{2}$, for 2nd (2nd order)

$$3-A_F = 1.848$$

$$A_F = 1.152$$

$$1 + \frac{R_f'}{R_1'} = 1.152$$

$$R_f' = 0.152 R_1'$$

$$\text{let } R_1' = 10 \text{ k}\Omega$$

$$R_f' = 1.52 \text{ k}\Omega$$

$$\text{let us assume, } R_2 = R_3 = R_4 = R_5 = R$$

$$C_2 = C_3 = C_4 = C_5 = C$$

$$f_h = \frac{1}{2\pi RC} = 10^3$$

$$\text{let, } C = 1 \mu\text{F}$$

$$R = \frac{1}{6.28 \times 10^3 \times 10^{-6}}$$

$$R = 0.159 \text{ k}\Omega$$

$$R = 159.23 \Omega$$

Difference b/w 1st & 2nd order LPF:

1. RC sections.

For 1st order - 1 RC section

For 2nd order - 2 RC sections.

2. Roll-off rate.

1st order = -20 dB/dec.

2nd order = -40 dB/dec.

For nth order LPF:

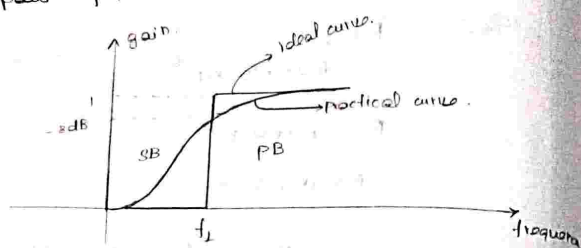
$$\left| \frac{V_o}{V_{in}} \right| = \frac{1}{\sqrt{1 + (f/f_h)^{2n}}} ; f_h = \frac{1}{2\pi RC}$$

$$\left| \frac{V_o(s)}{V_i(s)} \right| = \left| \frac{V_o(j\omega)}{V_i(j\omega)} \right| = |H(j\omega)|$$

$\rightarrow$ it allows low freq & it eliminates high frequency (Noise pH is used in band-limited signal)

HPF $\rightarrow$ It allows all the frequency above the cut-off frequency.

High pass filter:

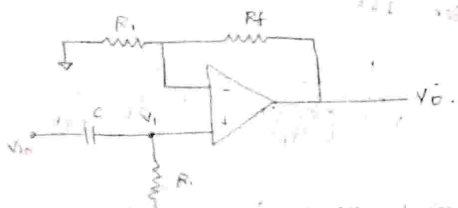


Exchange capacitor & Resistor in LPF

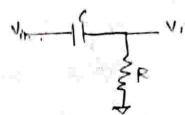
we get HPF

The filter which can allow all signal above the cutoff frequency and attenuate all the frequency signal below the cutoff frequency.

1st order active High pass filter.



$$V_o = \left(1 + \frac{R_f}{R_i}\right) V_i$$



$$V_i = \frac{V_{in}}{R + 1/sC} \cdot R$$

$$V_i = V_{in} \cdot \left(\frac{RSC}{1 + SRC} \right) \rightarrow (1)$$

$$V_o = \left(1 + \frac{R_f}{R_i}\right) V_i$$

$$= A_F \cdot \frac{V_{in}(SRC)}{1 + SRC}$$

$$\frac{V_o}{V_{in}} = A_F \cdot \frac{SRC}{1 + SRC}$$

$$\therefore S = j\omega$$

$$\omega = 2\pi f$$

$$\frac{V_o}{V_{in}} = A_F \cdot \left(\frac{j2\pi fRC}{1 + j2\pi fRC} \right)$$

$$\boxed{\frac{V_o}{V_{in}} = A_F \cdot \frac{j(f/f_L)}{1 + j(f/f_L)}}$$

$$\text{here, } f_L = \frac{1}{2\pi RC}$$

$f_L \rightarrow$ cut-off frequency of HPF.

case (i): $f \ll f_L$

$$\left| \frac{V_o(s)}{V_i(s)} \right| = \frac{A_F (f/f_L)}{\sqrt{1 + (f/f_L)^2}}$$

[It's gain is very less]

$$\left| \frac{V_o(s)}{V_i(s)} \right| < A_F \rightarrow \text{reduced.}$$

$\therefore$ It attenuates all the signal frequencies. $f \ll f_L$

case (ii): $f \gg f_1$

$$\left| \frac{V_o(s)}{V_i(s)} \right| > A_F$$

$$V_o(s) > A_F V_i(s)$$

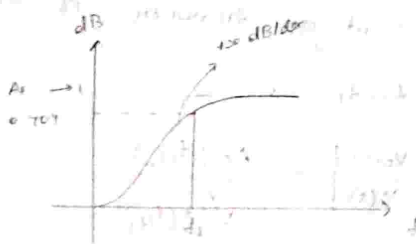
It allows all the signal frequency greater than cut-off frequency.

case (iii): $f = f_1$

$$\left| \frac{V_o(s)}{V_i(s)} \right| = \frac{A_F (f/f_1)}{\sqrt{1 + (f/f_1)^2}}$$

$$= \frac{A_F (1)}{\sqrt{1+1}} = \frac{A_F}{\sqrt{2}}$$

$$\left| \frac{V_o(s)}{V_i(s)} \right| = 0.707 A_F$$



Design a 1st HPF the cut off frequency of 2 kHz.

$$f_1 = 2 \text{ kHz}$$

$$f_1 = 2 \times 10^3 \text{ Hz}$$

$$f_1 = \frac{1}{2\pi RC} = 2 \times 10^3$$

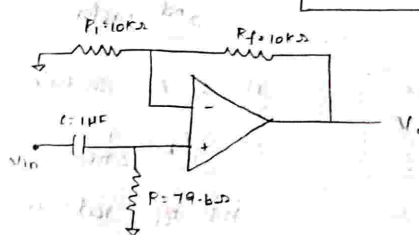
Let $C = 1 \mu\text{F}$,

$$R = \frac{1}{4 \times \pi \times 1 \times 10^{-6} \times 2 \times 10^3}$$

$$= \frac{1}{10.5663}$$

$$R = 0.094957 \text{ K}\Omega$$

$$R = 79.57 \Omega$$



$$A_F = 1 + \frac{R_f}{R_i} = 2$$

$$\frac{R_f}{R_i} = 1$$

$$= R_f = R_i$$

$$\text{Let us assume } R_i = R_f = 10 \text{ K}\Omega$$

Note:

* For n^{th} order high pass filter, the generalized expression of gain is

$$\left| \frac{V_o(s)}{V_i(s)} \right| = |H(s)| = \frac{A_F}{\sqrt{1 + (f_L/f)^{2n}}}$$

* For n^{th} order of LPF.

$$\left| \frac{V_o(s)}{V_i(s)} \right| = H(s) = \frac{A_F}{\sqrt{1 + (f/f_H)^{2n}}}$$

Difference b/w 1st & 2nd order HPF.

1st order

1. RC section

$$\Rightarrow f_L = \frac{1}{2\pi RC}$$

roll-off rate is

+20 db/dec.

2nd order

1. 2-RC section.

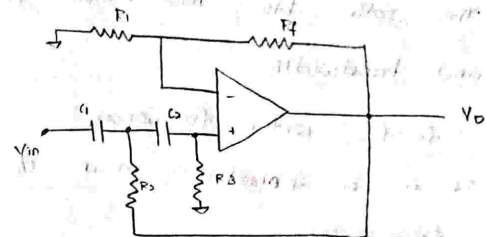
$$\Rightarrow f_H = \frac{1}{2\pi RC}$$

roll-off rate is

+40 db/dec.

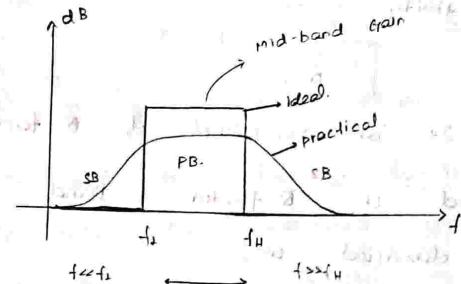
* For n^{th} order HPF. is +n20 db/dec.

2nd order high-pass filter:



Band-pass filter (BPF).

It allows all the signal frequency which is lies between two cut-off frequency. (f_L & f_H)



$\Rightarrow$ The signal is attenuated when $f \ll f_L$, $f \gg f_H$.

$\Rightarrow$ B/w f_L & f_H is called 'mid band frequency'.

Bandwidth:

* The range of frequency for which (the gain is maximum) or (the circuit is providing maximum gain)

* It is used to analyze the performance of filter.

Quality factor:

* The ratio b/w mid-band frequency (f_0) and bandwidth.

$\therefore f_0 = f_c = \text{center frequency.}$

$\Rightarrow$ It is a geometric mean of two cut-off frequencies.

$$Q = \frac{f_0}{BW} = \frac{f_c}{BW}$$

$$f_0 = f_c = \sqrt{f_L \cdot f_H}$$

selectivity:

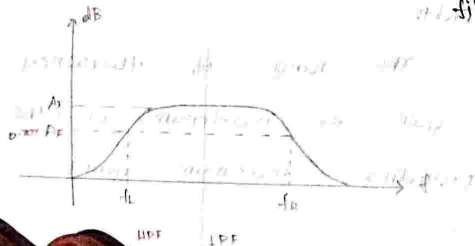
$$= \frac{1}{Q}$$

It is reciprocal of Q-factor.

• Based on Q-factor, Band pass filter is classified as,

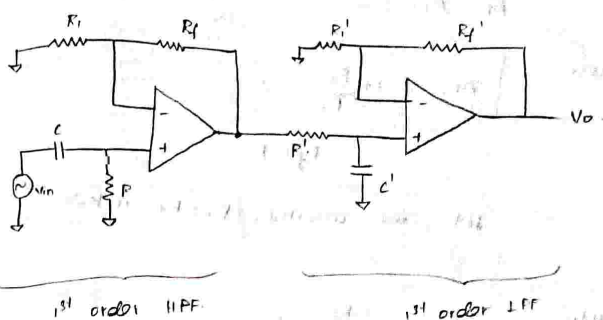
①. If $Q > 10 \rightarrow$ NBPF - Narrow band pass filter.

②. If $Q < 10 \rightarrow$ WBPF - wide band pass filter.



1st order wide Band Pass filter (WBPF).

* The HPF is cascaded with LPF.



$$\left| \frac{V_o(s)}{V_{in}(s)} \right| = |H(s)| = \frac{A_{F1} \left(\frac{f}{f_L} \right)}{\sqrt{1 + \left(\frac{f}{f_L} \right)^2}} \cdot \frac{A_{F2}}{\sqrt{1 + \left(\frac{f}{f_H} \right)^2}}$$

$\therefore$ gain of WBPF = Product of HPF & LPF

②. Design a wide BPF having $f_L = 400 \text{ Hz}$ and $f_H = 2 \text{ kHz}$ and pass band gain of 4. also find the quality factor of the filter circuit.

Sol:

$$f_L = 400 \text{ Hz}$$

$$f_H = 2 \text{ kHz}$$

WBPF gain of 4.

$$Q = ?$$

$$A_1 = A_{F1} - A_{F2}$$

$$A_{F1} = A_{F2} = 0$$

here, $A_{F1} = 1 + \frac{R_f}{R_i} = 0$

$$R_f = R_i$$

let us assume, $R_i = R_f = 10 \text{ k}\Omega$

also, $A_{F2} = 1 + \frac{R_f'}{R_i'} = 0$

$$\frac{R_f'}{R_i'} = -1$$

$$R_f' = R_i'$$

let us assume, $R_i' = R_f' = 10 \text{ k}\Omega$

for HPF, $f_c = 400 \text{ Hz}$

$$f_c = \frac{1}{2\pi RC} = 400$$

let $C = 1 \mu\text{F}$

$$R = \frac{1}{400 \times 2\pi \times 10^{-6}}$$

$$= \frac{1}{2513.27} \times 10^6$$

$$= 397.88 \times 10^3$$

$$R = 397.88 \text{ k}\Omega$$

for LPF, $f_H = 5 \text{ kHz}$

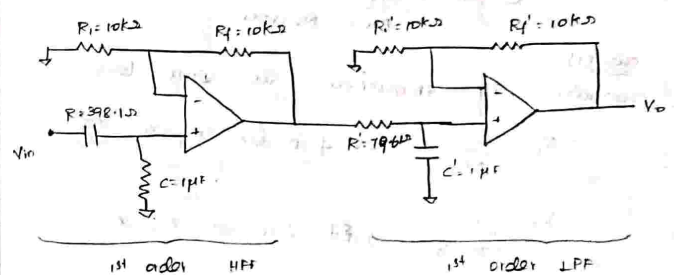
$$f_H = \frac{1}{2\pi R' C'} = 5 \times 10^3$$

let $C = 1 \mu\text{F}$

$$R' = \frac{1}{4 \times \pi \times 10^3 \times 1 \times 10^{-6}}$$

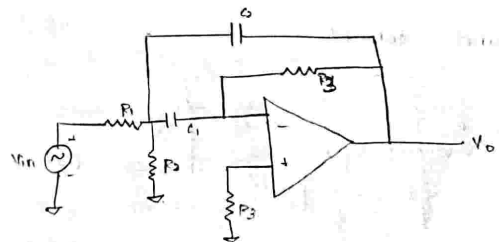
$$= 0.07961 \times 10^3$$

$$R' = 79.61 \Omega$$



Narrow Band pass filter: (NBPF)

⇒ The input is given in inverting terminal.



① op-amp is configured in inverting mode.

② It has two - feed back.

because of this feedback it is named as multiple feedback filter.

③. C_1 is responsible for make this circuit fn as LPF.

C_2 is responsible for make this circuit fn as HPF.

(H). $Q > 10 = \frac{f_0}{\text{BW}} \leftarrow \text{Narrow}$

case (i):

consider i/p frequency is very less.

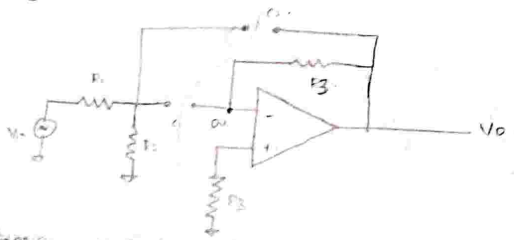
$$X_{C1} = \frac{1}{2\pi f C_1}, \text{ if } f=0 \Rightarrow X_{C1} = \infty.$$

$$X_{C2} = \frac{1}{2\pi f C_2}, \text{ if } f=0 \Rightarrow X_{C2} = \infty.$$

$\therefore$ If $X_{C1} = X_{C2} = \infty$ it act as open circuit.

$V_0 = 0$ due to virtual

ground concept.

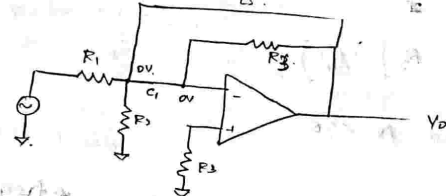


$\Rightarrow$ It exhibits the behaviour of HPF.

case (ii):

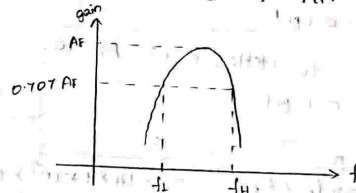
consider $f_{in} > f_H$

$X_{C1} \& X_{C2} = 0$ [it act as closed switch/circuit].
 $\Rightarrow$ It exhibits the behaviour of LPF.



case (iii):

consider, $f_L < f < f_H$.



advantage:

* By changing R_3 we can transient the center frequency of NBPF from one value to another value.

steps: Design parameters for NBPF:

1). assume $C_1 = C_2 = C$.

2). $R_1 = \frac{0}{2\pi f_0 C AF} = \frac{0}{2\pi f_c \cdot C \cdot AF}$

$$3). R_2 = \frac{Q}{2\pi f_0 C [2Q^2 - AF]}$$

$$4). R_3 = \frac{Q}{\pi f_0 C}$$

$$5). A_F = \frac{R_3}{2R_1}$$

$$6). R_3' = R_3 \left[\frac{f_0}{f_0'} \right]$$

also $AF < 2Q^2$.

$f_0 \rightarrow$ initial value of centre frequency
change to

$f_0' \rightarrow$ final value
change $R_3 \rightarrow R_3'$

11. Design a NBPB with centre frequency of 1 kHz, $Q = 13$ and $AF = 10$ also assume $C = 0.1 \mu F$.

sol: Given, $f_0 = 1 \text{ kHz}$, $Q = 13$, $AF = 10$.

1). $C_1 = C_2 = 0.1 \mu F$.

$$2). R_1 = \frac{Q}{2\pi f_0 C AF} = \frac{13}{2 \times 3.14 \times 1 \times 10^3 \times 10^{-6} \times 10} = \frac{13}{6.28}$$

$$R_1 = 2.07 \text{ k}\Omega$$

$$3). R_2 = \frac{Q}{2\pi f_0 C [2Q^2 - AF]} = \frac{13}{6.28 \times 10^3 \times 10^{-6} [2 \times 169 - 10]} = \frac{13}{6.28 \times (338) \times 10^{-4}} = \frac{13}{2079.84} \times 10^4 = 6.3111 \times 10^4 \times 10^{-3}$$

$$R_2 = 63.111 \Omega$$

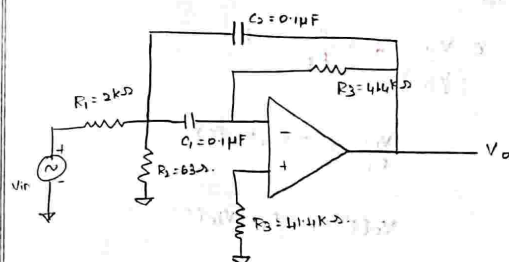
$$4). R_3 = \frac{Q}{\pi f_0 C} = \frac{13}{3.14 \times 1 \times 10^3 \times 10^{-6}} = 4.1401 \times 10^4$$

$$R_3 = 41.401 \text{ k}\Omega$$

$$5). A_F = \frac{R_3}{2R_1} = \frac{41.401 \times 10^3}{2 \times 2.07 \times 10^3} = \frac{41.401}{4.14}$$

$$= 10.000$$

$$AF = 10$$



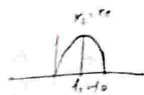
Find R_3' from $f_0 = 1 \text{ kHz}$ to 10 kHz .

$$R_3' = R_3 \left(\frac{f_0}{f_0'} \right)$$

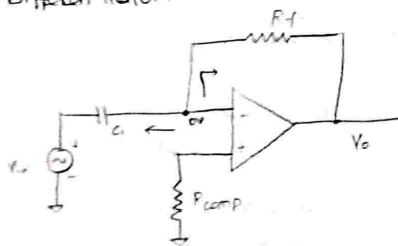
$$= 63.111 \times \frac{1 \times 10^3}{10 \times 10^3}$$

$$R_3' = 6.311 \Omega$$

Resonance - at which the gain is max
($V_2 = x_c$)



Differentiator:



$\Rightarrow$ It differentiates the applied i/p.

Apply Nodal equation,

$$\frac{0 - V_{in}}{1/sC_i} + \frac{0 - V_o}{R_f} = 0$$

$$\frac{V_o(s)}{R_f} = -sC_i V_{in}(s)$$

$$V_o(s) = -sR_f C_i V_{in}(s)$$

$$\frac{V_o(s)}{V_{in}(s)} = -R_f C_i s$$

Inverse Laplace of V_o

$$\mathcal{L}^{-1}[V_o(s)] \Rightarrow V_o(t) = -R_f C_i \frac{dV_{in}(t)}{dt}$$

* Expression of ideal differentiator.

$$\text{Gain, } \frac{V_o(s)}{V_{in}(s)} = -R_f C_i (j\omega)$$

$$\left| \frac{V_o(s)}{V_{in}(s)} \right| = R_f C_i \omega$$

$$= R_f C_i 2\pi f$$

$$\left| \frac{V_o(s)}{V_{in}(s)} \right| = \frac{f}{f_0}$$

$$\text{where, } f_0 = \frac{1}{2\pi R_f C_i}$$

* Gain expression of ideal differentiation.

Limitation of ideal differentiator:

$\Rightarrow$ when input signal frequency $\uparrow$ so what (about)

(i) the $\rightarrow$ i/p impedance.
 $\searrow$ gain.

* i/p frequency $\uparrow$ so - gain is more, it leads unstable.

* for high i/p frequency - i/p impedance is very low bcoz of this the circuit will be affected by high frequency noise.

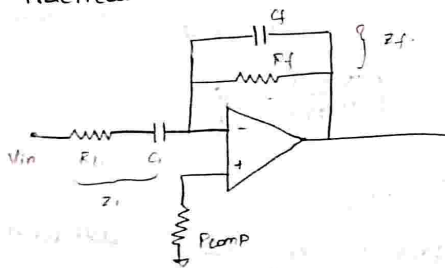
$\Rightarrow$ To overcome this we add

one Resistor, so that the minimum impedance is 'R' when high frequency.

... .. limit free noise.

⇒ to improve the stability we add one capacitor in feedback.

Practical differentiator:



$$Z_f = R_f \parallel \frac{1}{sC_f}, \quad Z_i = R_1 + \frac{1}{sC_1}$$

$$\frac{V_o}{V_{in}} = -\frac{Z_f}{Z_i}$$

$$Z_i = \frac{sR_1C_1 + 1}{sC_1}$$

$$\frac{1}{Z_f} = \frac{1}{R_f} + \frac{1}{1/sC_f}$$

$$= \frac{1}{R_f} + sC_f$$

$$= \frac{1 + sR_fC_f}{R_f}$$

$$Z_f = \frac{R_f}{1 + sR_fC_f}$$

$$\frac{V_o}{V_{in}} = -\frac{R_f}{\frac{1 + sR_fC_f}{sR_1C_1}} = -\frac{sR_1C_1R_f}{1 + sR_fC_f}$$

$$\frac{V_o}{V_{in}} = \frac{-sR_fC_1}{(1 + sR_fC_f)(1 + sR_1C_1)}$$

$$\left| \frac{V_o}{V_{in}} \right| = \frac{sR_fC_1}{(1 + sR_fC_f)(1 + sR_1C_1)}$$

assume, $R_fC_f = R_1C_1$.

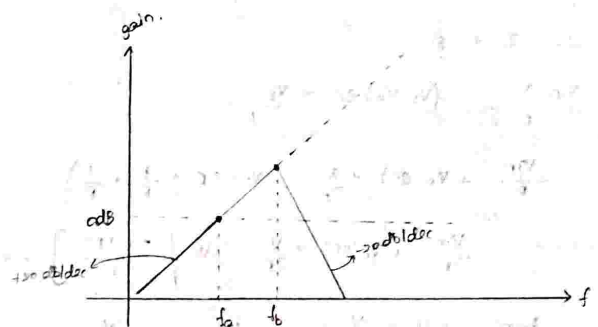
$$\left| \frac{V_o}{V_{in}} \right| = \frac{sR_fC_1}{[1 + sR_1C_1]^2}$$

$$s = j\omega = j2\pi f$$

$$= \frac{sR_fC_1}{[1 + j2\pi fR_1C_1]^2}$$

$$\left| \frac{V_o}{V_{in}} \right| = \frac{sR_fC_1}{[1 + j(f/f_b)]^2}$$

$$\text{here, } f_b = \frac{1}{2\pi R_1C_1} = \frac{1}{2\pi R_fC_f}$$



$$f = f_a, \quad A = 1$$

$$A = 0 \text{ dB}$$

conditions to design:

→ $f_b = 10 f_a$

→ signal frequency $f_s = \frac{f_a}{20}$

→ $T \geq R_f C_1$

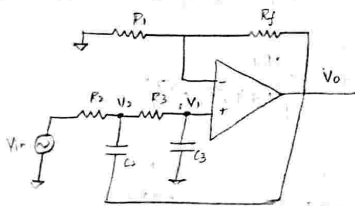
→ $R_1 C_1 = R_f C_f$

→ use f_a find R_f

→ use $f_b = 10 f_a$ find f_b

→ use f_b find R_1

2nd order low pass filter:



$$V_o = \left[1 + \frac{R_f}{R_1} \right] V_1 \quad \text{--- (1)}$$

Let, $R_2 = R_3 = R$, $C_2 = C_3 = C$

$$I_1 = I_2 + I_3$$

$$\frac{V_{in} - V_2}{R} = (V_2 - V_o) sC + \frac{V_2 - V_1}{R}$$

$$\frac{V_{in}}{R} + V_o(sC) + \frac{V_1}{R} = V_2 \left(sC + \frac{1}{R} + \frac{1}{R} \right)$$

$$\frac{V_{in}}{R} + V_o(sC) + \frac{V_1}{R} = V_2 \left[\frac{2 + sRC}{R} \right] \quad \text{--- (2)}$$

$$\text{Now, } V_1 = \frac{V_2}{R + 1/sC} = \frac{V_2}{1 + sRC}$$

$$\therefore V_2 = V_1 (1 + sRC)$$

apply this into equ (2)

$$\frac{V_{in}}{R} + V_o(sC) + \frac{V_1}{R} = \frac{V_1 (1 + sRC) (2 + sRC)}{R}$$

$$\frac{V_{in}}{R} + V_o(sC) = \frac{V_1 [2 + sRC + 2sRC - (sRC)^2] - V_1}{R}$$

$$\frac{V_{in}}{R} + V_o(sC) = \frac{V_1}{R} [2 + 3sRC + (sRC)^2 - 1]$$

$$\frac{V_{in}}{R} + V_o(sC) = \frac{V_1}{R} [1 + 2sRC + (sRC)^2]$$

$$V_{in} + V_o(sRC) = V_1 [1 + 2sRC + (sRC)^2]$$

$$\therefore V_1 = \frac{V_{in} + V_o(sRC)}{1 + 2sRC + (sRC)^2}$$

apply this into equ (1)

$$V_o = A_f \left[\frac{V_{in} + V_o(sRC)}{1 + 2sRC + (sRC)^2} \right]$$

$$V_o = A_f \left[\frac{V_{in}}{1 + 2sRC + (sRC)^2} \right] + A_f \left[\frac{V_o(sRC)}{1 + 2sRC + (sRC)^2} \right]$$

$$V_o \left[1 - \frac{A_f(sRC)}{1 + 2sRC + (sRC)^2} \right] = A_f \frac{V_{in}}{1 + 2sRC + (sRC)^2}$$

$$V_o \left[\frac{1 + 2sRC + (sRC)^2 - A_f(sRC)}{1 + 2sRC + (sRC)^2} \right] = A_f \frac{V_{in}}{1 + 2sRC + (sRC)^2}$$

$$\therefore \frac{V_o}{V_{in}} = \frac{A_f}{1 + sRC [2 - A_f] + (sRC)^2}$$

$$\begin{aligned}\frac{V_o}{V_{in}} &= \frac{AF}{1 + sRC[3 - AF] + s^2 R^2 C^2} \\ &= \frac{AF}{R^2 C^2 \left[s^2 + \frac{s}{RC} (3 - AF) + \frac{1}{R^2 C^2} \right]} \\ &= \frac{AF / (RC)^2}{s^2 + \left[\frac{3 - AF}{RC} \right] s + \frac{1}{R^2 C^2}} \\ &= \frac{AF \omega_h^2}{s^2 + \alpha \omega_h s + \omega_h^2} \quad \left(\begin{array}{l} \omega_h = \frac{1}{RC} \\ \alpha = 3 - AF \end{array} \right)\end{aligned}$$

$$\frac{V_o}{V_{in}} = AF \cdot \frac{1/R^2 C^2}{s^2 + \left[\frac{3 - AF}{RC} \right] s + \frac{1}{R^2 C^2}} \rightarrow (3)$$

we know that for 2nd order system.

$$\frac{V_o}{V_{in}} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{1}{s^2 + \sqrt{2}s + 1} \rightarrow (4)$$

compare (3) & (4).

$$\frac{3 - AF}{RC} = \sqrt{2}$$

$$\text{Let } RC = 1, \quad 3 - AF = \sqrt{2}$$

$$AF = 3 - \sqrt{2}$$

$$AF = 1.586$$

$$\text{For } n^{\text{th}} \text{ order LPT, } \left| \frac{H(j\omega)}{A_0} \right| = \frac{1}{\sqrt{1 + (\omega/\omega_h)^{2n}}}$$

$$\text{Also, } \omega_h^2 = \omega_0^2 = \frac{1}{R^2 C^2}$$

$$\omega_0 = \frac{1}{RC}$$

$$f_0 = \frac{1}{2\pi RC}$$

$$\text{If } R_1 = R_2 = R, \quad C_1 = C_2 = C$$

$$f_0 = \frac{1}{2\pi \sqrt{R_1 R_2 C_1 C_2}}$$

design a second order butterworth lowpass filter having upper cut-off frequency of 1 kHz.

Sol:

$$AF = 1 + \frac{R_f}{R_1} = 1.586$$

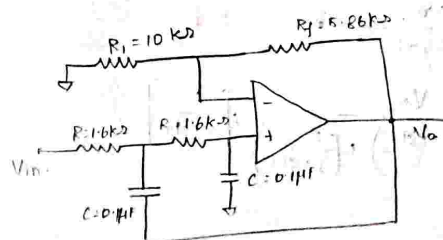
$$R_f = 0.586 R_1$$

$$\text{Let } R_1 = 10 \text{ k}\Omega, \quad R_f = 5.86 \text{ k}\Omega$$

$$f_0 = \frac{1}{2\pi RC} = 1000$$

$$\text{Let } C = 0.1 \mu\text{F}, \quad R = \frac{1}{2\pi \times 0.1 \times 10^{-6} \times 1000}$$

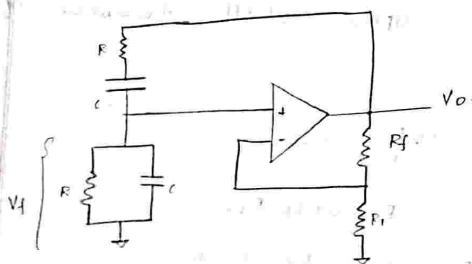
$$R = 1.6 \text{ k}\Omega$$



order factors of polynomials.

1. $s+1$
2. $s^2 + \sqrt{2}s + 1$
3. $(s+1)(s^2+s+1)$
4. $(s^2+0.765s+1)(s^2+1.848s+1)$
5. $(s+1)(s^2+0.618s+1)(s^2+1.618s+1)$

Wien- Bridge oscillator.



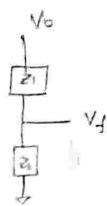
$$\text{Let } Z_1 = R + \frac{1}{sC} = \frac{1+sRC}{sC}$$

$$Z_2 = R \parallel \frac{1}{sC} = \frac{R \times 1/sC}{R + 1/sC} = \frac{R}{sRC + 1}$$

here, V_f = Voltage across ' Z_2 '

$$V_f = \frac{V_o}{Z_1 + Z_2} \cdot Z_2$$

$$= \frac{V_o}{\left(\frac{1+sRC}{sC}\right) + \left(\frac{R}{1+sRC}\right)} \cdot \left[\frac{R}{1+sRC}\right]$$



$$= \frac{V_o (1+sRC)(sC)}{\left[\left(\frac{1+sRC}{sC}\right) + \frac{R}{(1+sRC)}\right]} \cdot \frac{R}{(1+sRC)}$$

$$V_f = \frac{V_o (sRC)}{1 + (sRC)^2 + sRC + sRC}$$

$$V_f = \frac{V_o (sRC)}{1 + 3sRC + (sRC)^2}$$

$$\therefore \beta = \frac{V_f}{V_o} = \frac{sRC}{1 + 3sRC + (sRC)^2}$$

$$= \frac{sRC}{sRC \left[3 + sRC + \frac{1}{sRC} \right]}$$

$$= \frac{1}{3 + j\omega RC + \frac{1}{j\omega RC}}$$

$$\beta = \frac{1}{3 + j \left[\omega RC - \frac{1}{\omega RC} \right]}$$

to have sustained oscillations ' β ' must be a Real.

$$\beta = \frac{V_f}{V_o} = \frac{1}{3}$$

equating imaginary part to '0'.

$$\omega RC - \frac{1}{\omega RC} = 0$$

$$\omega^2 R^2 C^2 = 1$$

$$\omega^2 = \frac{1}{R^2 C^2} \Rightarrow \omega = \frac{1}{RC}$$

$$f_0 = \frac{1}{2\pi RC}$$

To satisfy the magnitude condition,

$$A_v \beta = 1.$$

$$A_v \left(\frac{1}{3} \right) = 1.$$

$$A_v = 3.$$

(3) Design a Wien bridge oscillator for

$$f_0 = 1 \text{ kHz}.$$

Sol: Given, $f_0 = 1 \text{ kHz} = 1000 \text{ Hz}.$

W.K.T $f_0 = \frac{1}{2\pi RC} = 1000$

Let $C = 0.1 \mu\text{F}.$

$$R = \frac{1}{2\pi \times 1000 \times 0.1 \times 10^{-6}}$$

$$R = 1.59 \text{ k}\Omega.$$

$$R \approx 1.6 \text{ k}\Omega.$$

for Wien-Bridge

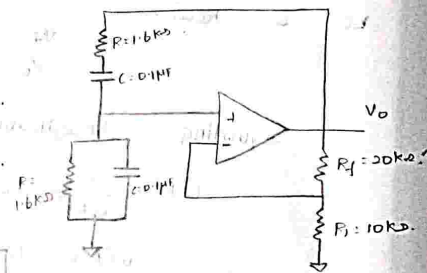
$$A_v = 3.$$

$$\left[1 + \frac{R_f}{R_1} \right] = 3.$$

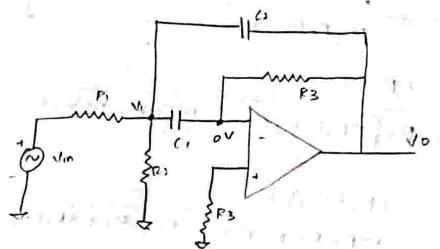
$$R_f = 2R_1.$$

Let $R_1 = 10 \text{ k}\Omega.$

$$\therefore R_f = 20 \text{ k}\Omega.$$



Narrow Band Pass Filter (NBPF).



Let, $Z_1 = 1/sC_1$, $Z_2 = 1/sC_2$

write Nodal equation w.r.t Node 'V1'

$$\frac{V_1 - V_{in}}{R_1} + \frac{V_1}{R_2} + (V_1 - V_o) sC_2 + V_1 (sC_1) = 0.$$

$$V_1 \left[\frac{1}{R_1} + \frac{1}{R_2} + sC_2 + sC_1 \right] - \frac{V_{in}}{R_1} - V_o (sC_2) = 0 \rightarrow (1)$$

write Nodal eqn w.r.t Node 'O'

$$(0 - V_1) sC_1 + \frac{(0 - V_o)}{R_3} = 0.$$

$$\frac{V_o}{R_3} = -V_1 sC_1$$

$$V_1 = \frac{-V_o}{sR_3C_1} \rightarrow (2)$$

apply equation (2) into equation (1)

$$\left[-\frac{V_o}{sR_3C_1} \right] \left[\frac{1}{R_1} + \frac{1}{R_2} + sC_2 + sC_1 \right] - \frac{V_{in}}{R_1} - V_o (sC_2) = 0.$$

$$V_o (sC_2) + \frac{V_o}{sR_3C_1} \left[\frac{1}{R_1} + \frac{1}{R_2} + sC_2 + sC_1 \right] = -\frac{V_{in}}{R_1}$$

$$V_o \left[sC_2 + \frac{1}{sR_3C_1} \left(\frac{R_2 + R_1 + sR_1R_2C_2 + sR_1R_2C_1}{R_1R_2} \right) \right] = -\frac{V_{in}}{R_1}$$

$$V_o \left[\frac{s^2 R_1 R_2 R_3 C_1 C_2 + R_2 + R_1 + s R_2 R_1 C_2 + s R_1 R_2 C_1}{s R_1 R_2 R_3 C_1} \right] = -\frac{V_{in}}{R_1}$$

$$\frac{V_o}{V_{in}} = \frac{-s R_1 R_2 R_3 C_1}{R_1 [s^2 R_1 R_2 R_3 C_1 C_2 + R_2 + R_1 + s R_2 R_1 C_2 + s R_1 R_2 C_1]}$$

$$\frac{V_o}{V_{in}} = \frac{-(R_2 R_3 C_1) s}{s^2 R_2 R_3 R_1 C_1 C_2 + R_2 + R_1 + s R_2 R_1 C_2 + s R_1 R_2 C_1}$$

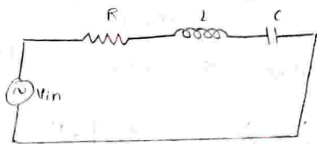
$$\frac{V_o}{V_{in}} = \frac{-(R_2 R_3 C_1) s}{R_1 R_2 R_3 C_1 C_2 \left[s^2 + \frac{1}{R_1 R_3 C_1 C_2} + \frac{1}{R_2 R_3 C_1 C_2} + \frac{s}{R_3 C_1} + \frac{s}{R_1 C_2} \right]}$$

Let, $C_1 = C_2 = C$

$$\frac{V_o}{V_{in}} = \frac{-(s/R_1 C_2)}{s^2 + \frac{1}{R_1 R_3 C^2} + \frac{1}{R_2 R_3 C^2} + \frac{2s}{R_3 C}}$$

$$\therefore \frac{V_o}{V_{in}} = \frac{(-s/R_1 C)}{s^2 + \frac{2s}{R_3 C} + \frac{1}{R_3 C^2} \left[\frac{1}{R_2} + \frac{1}{R_1} \right]} \rightarrow (3)$$

Transfer fn of RLC series resonant circuit.



Overall impedance.

$$Z = R + sL + \frac{1}{sC} = \frac{sRC + s^2 LC + 1}{sC}$$

$$V_o = \frac{V_{in} \cdot AF}{\left[\frac{sRC + s^2 LC + 1}{sC} \right]}$$

AF $\rightarrow$ Max Gain at resonance

$$= \frac{V_{in} (sRC) \cdot AF}{LC \left[s^2 + \frac{R}{L} s + \frac{1}{LC} \right]}$$

Note:
 $\Rightarrow$ at resonance Max
o/p occurs
across the resistor
'R'.

$$V_o = \frac{V_{in} (R/L) s \cdot AF}{LC \left[s^2 + (R/L) s + 1/LC \right]}$$

$$\therefore \frac{V_o}{V_{in}} = \frac{-s (R/L) \cdot AF}{s^2 + \left(\frac{R}{L} \right) s + \frac{1}{LC}} \rightarrow (4)$$

$\therefore V_o = V_R$

here, Quality factor, $Q = \frac{\omega_0 L}{R}$

$$\therefore \frac{R}{L} = \frac{\omega_0}{Q} \rightarrow (5)$$

$$\text{Also } \omega_0^2 = \frac{1}{LC}$$

$$\omega_0 = \frac{1}{\sqrt{LC}} \rightarrow (6)$$

apply equ (5), (6) into equ (4).

$$\therefore \frac{V_o}{V_{in}} = \frac{-s (\omega_0/Q) \cdot AF}{s^2 + \left(\frac{\omega_0}{Q} \right) s + \omega_0^2} \rightarrow (7)$$

The above equation, (7) represents the transfer fn of RLC series resonant circuit.

Compare equ (7) with equ (3).

$$AF \cdot \frac{\omega_0}{Q} = \frac{1}{R_1 C}$$

$$\therefore R_1 = \frac{Q}{\omega_0 C \cdot AF} = \frac{Q}{2\pi f_0 C \cdot AF} \rightarrow (8)$$

$$\text{Hence } \frac{\omega_0}{Q} = \frac{2}{R_3 C}$$

$$R_3 = \frac{2Q}{\omega_0 C} = \frac{2Q}{2\pi f_0 C} = \frac{Q}{\pi f_0 C}$$

$$R_3 = \frac{Q}{\pi f_0 C} \rightarrow (9)$$

$$\text{Also, } \omega_0^2 = \frac{1}{R_3 C^2} \left[\frac{1}{R_2} + \frac{1}{R_1} \right]$$

$$\omega_0^2 [R_3 C^2] = \frac{1}{R_2} + \frac{1}{R_1}$$

$$\frac{1}{R_2} = \omega_0^2 [R_3 C^2] - \frac{1}{R_1}$$

$$\frac{1}{R_2} = \omega_0^2 [R_3 C^2] - \frac{1}{(Q/2\pi f_0 C A_F)}$$

$$= \omega_0^2 R_3 C^2 - \frac{2\pi f_0 C A_F}{Q}$$

$$\frac{1}{R_2} = \frac{Q\omega_0^2 [R_3 C^2] - 2\pi f_0 C A_F}{Q}$$

$$\therefore R_2 = \frac{Q}{Q\omega_0^2 (R_3 C^2) - 2\pi f_0 C A_F}$$

$$= \frac{Q}{Q\omega_0^2 \left[\frac{Q}{\pi f_0 C} \right] C^2 - 2\pi f_0 C A_F}$$

$$= \frac{Q}{Q\omega_0^2 \left[\frac{2QC^2}{2\pi f_0 C} \right] - 2\pi f_0 C A_F}$$

$$= \frac{Q}{2Q^2\omega_0 C - 2\pi f_0 C A_F} \quad \because \omega_0 = 2\pi f_0$$

$$= \frac{Q}{\omega_0 C [2Q^2 - A_F]}$$

$$\therefore R_2 = \frac{Q}{2\pi f_0 C [2Q^2 - A_F]} \rightarrow (10)$$

from equ (9).

$$f_0 = \frac{Q}{\pi R_3 C} \rightarrow (11)$$

Hence from equ (2).

$$f_0 = \frac{Q}{2\pi R_1 C A_F} \rightarrow (12)$$

equating (11) & (12).

$$\frac{Q}{\pi R_3 C} = \frac{Q}{2\pi R_1 C A_F}$$

$$\therefore A_F = \frac{R_3}{2R_1} \rightarrow (13)$$

to change centre frequency from $f_0 \rightarrow f_0'$

$$\therefore R_2' = R_2 \left[\frac{f_0}{f_0'} \right]^2$$